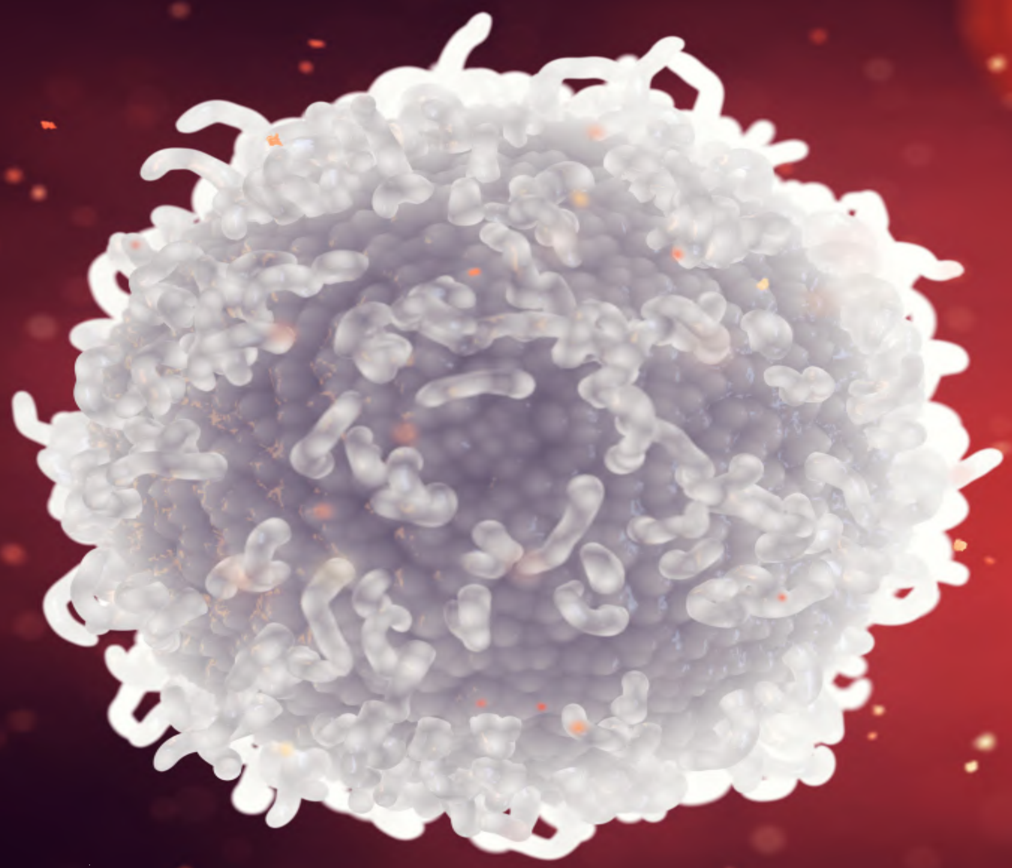


LEUKOCYTES

Leukopoiesis : formation of WBC

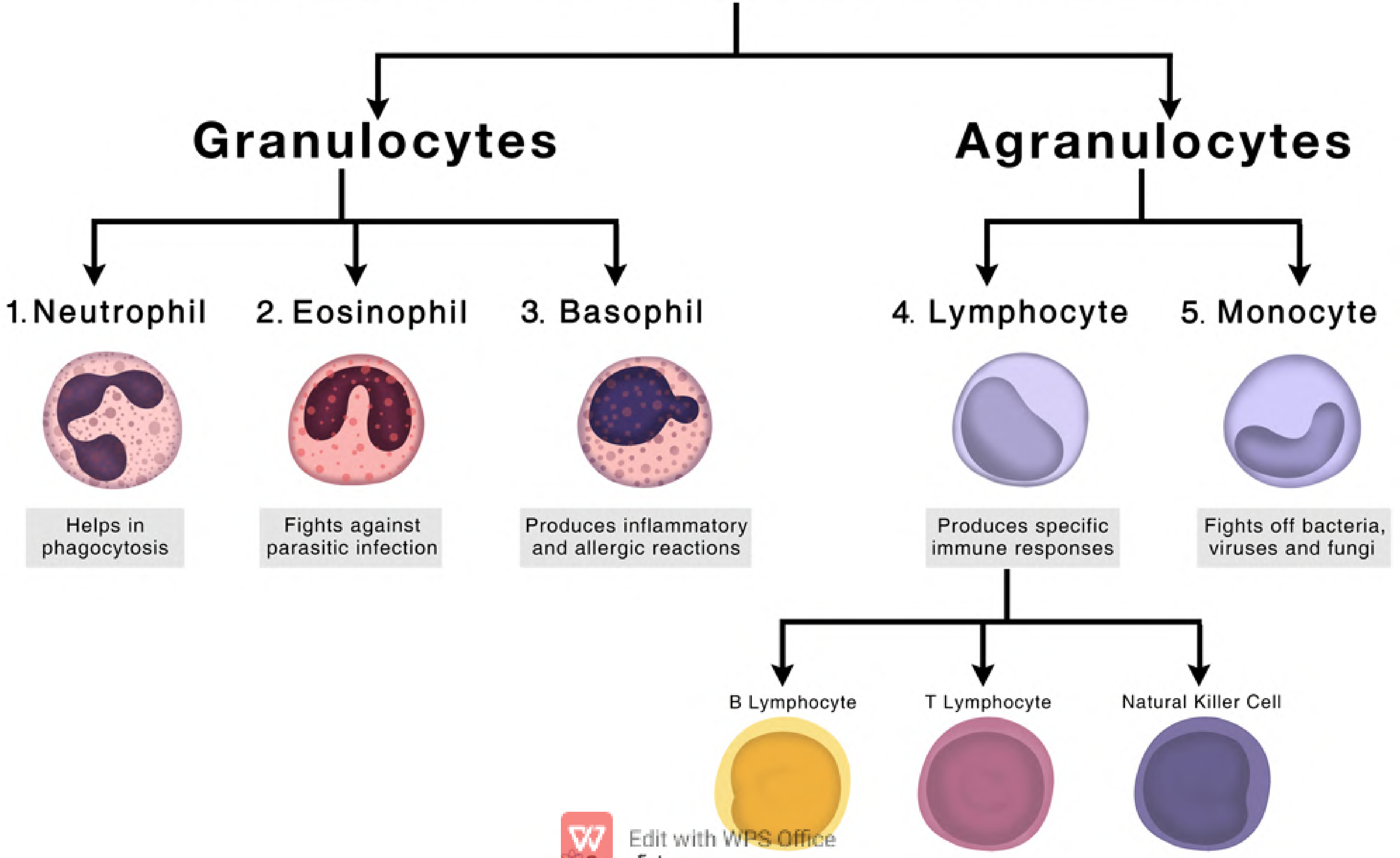
- **Colourless cells**
- Only blood cells with complete organelles like nucleus, mitochondria, etc.
- **Less numerous** than RBCs :- 4000 - 11000 /cu mm
- **Defend the body** against diseases by fighting infections, antigens and also against malignancy
- Normal life span of WBCs is **7 – 10 days.**

Classification

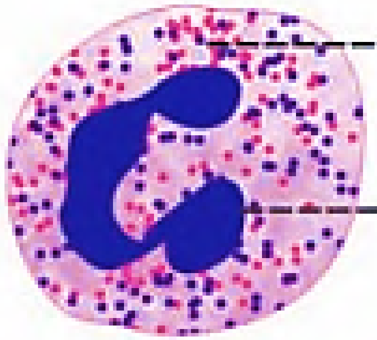
- Based on the presence or absence of granules in the cytoplasm, the leukocytes are classified into two groups:

1. **Granulocytes** which have granules.
2. **Agranulocytes** which do not have granules.

TYPES OF WHITE BLOOD CELLS

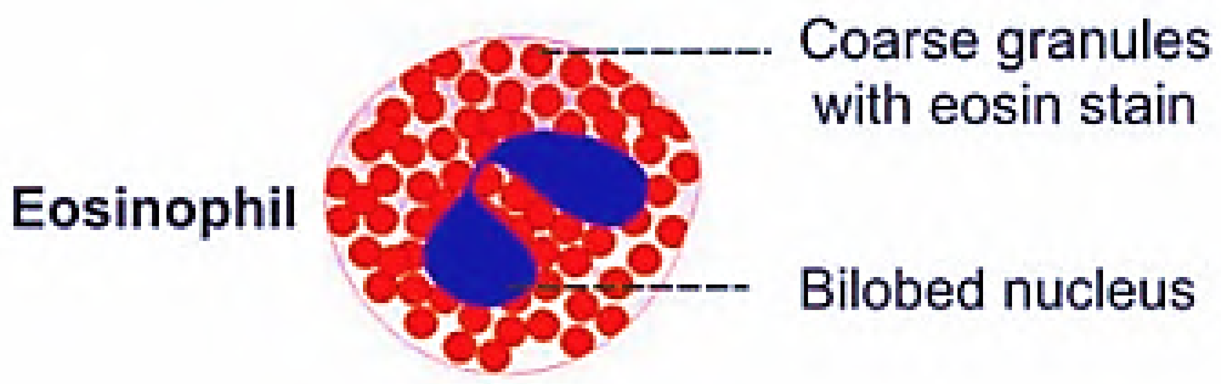


Neutrophil:



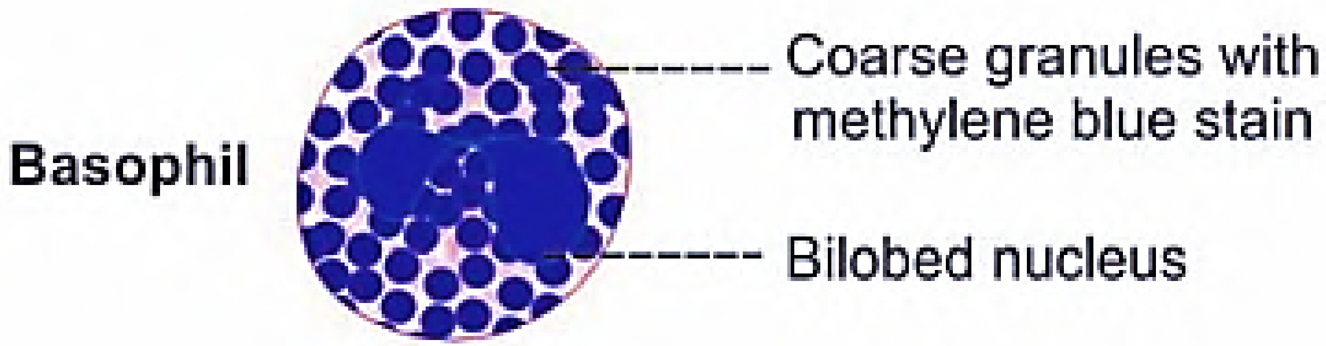
- Numerous: **50-70%** of the total WBCs.
- Also called **Polymorphonuclear leukocytes.**
- **Multi-lobed nucleus** 2-6 lobes.
- Contains both **acidic and basic granules.**

Eosinophil:



- 2-4% of the total WBCs.
- Bilobed nucleus with the shape of a phone receiver.
- Acidic granules.

Basophil:

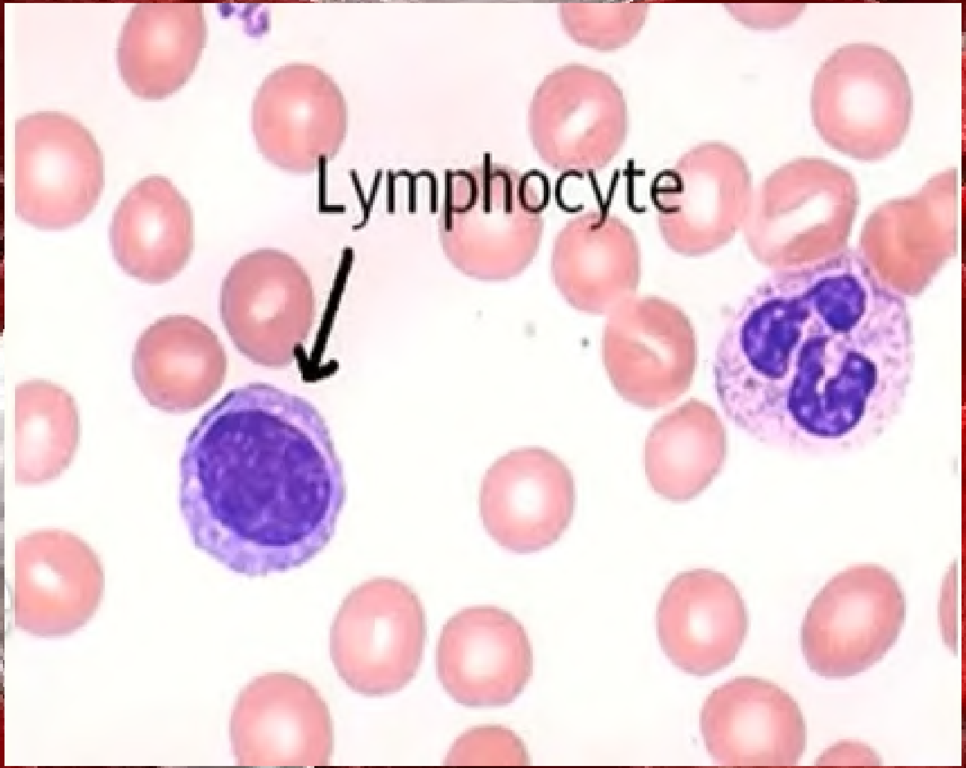


- 0-1% of the total WBCs.
- Bilobed nucleus, ‘S’ or ‘V’ shaped but masked by granules.
- Basic granules.

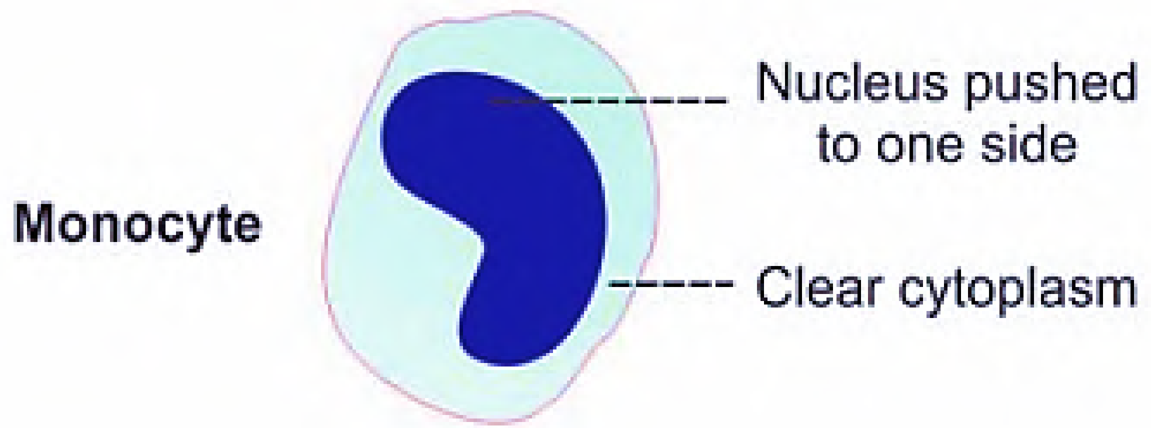
Lymphocytes:

- 20-30%
- 2 Types: -
- Small lymphocytes and large lymphocytes
- Agranulated, round nucleus


SL


Lymphocyte

Monocytes:

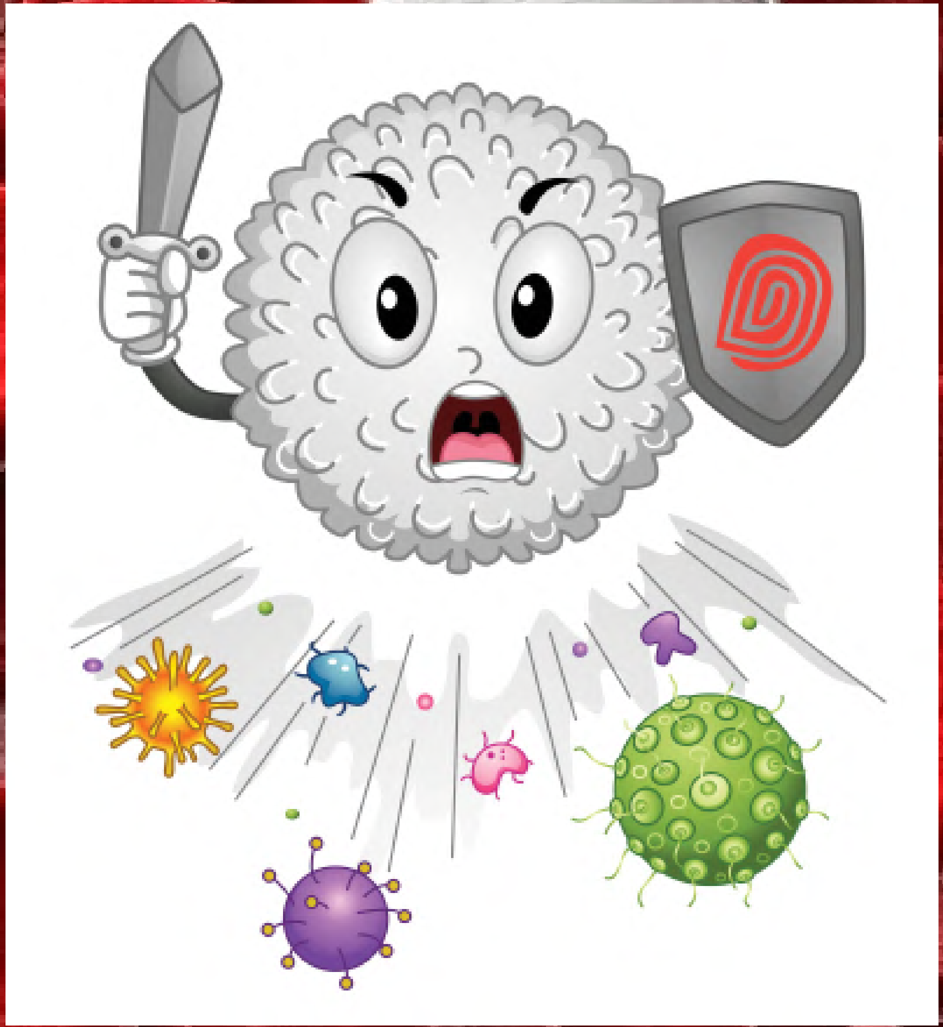


- 2-6%
- Largest of all WBCs.
- Kidney shaped nucleus.

Properties of WBC

- **Diapedesis:** It is the process by which the leukocytes squeeze through the normal capillaries.
- **Amoeboid movement:** By protruding in to the cytoplasm and changing the shape.
- **Chemotaxis:** A number of chemical substances in the tissue causes leukocytes move toward tissues.
- **Phagocytosis:** **Neutrophils** and **monocytes** swallow foreign body by means of pseudopodia

FUNCTIONS:



Functions:

- **Neutrophils**- 1st line of defence cells- phagocytosis.
- **Eosinophils**- release cytotoxic substances from the granules & Responsible for detoxification, disintegration & removal of foreign body.
- **Basophils**- healing process after inflammation and hypersensitivity reactions- releases histamine and other chemicals.
- **Lymphocytes**- functionally 2 types- **B cells** (humoral immunity) & **T cells** (cell mediated immunity).
- **Monocytes** – 2^{nd} line of defence cells – phagocytosis.

VARIATIONS

Physiological

- Age- high in infants
- Gender- more in males, increases during menstruation, pregnancy & parturition in females
- Increases during exercise and emotions

Pathological

- Leukopenia- decreased WBC
- Leukocytosis- Increased WBC
- Leukemia- Uncontrolled increase in WBC

Physiological variations:

1. **Age:** In infants, count is about 20,000 mm3 and in children it is about 10,000- 15,000 mm3.
2. **Gender:** More in males than in females. In females it increases during menstruation, pregnancy and parturition.
3. **Diurnal variation:** Minimum in early morning and maximum in after noon.
4. **Exercise:** Increase in count.
5. **Sleep:** Minimum count.
6. **Emotional conditions:** It increases during emotional conditions like anxiety.

Pathological variations:

- **Leukopenia:** Decrease in WBC count. It occurs during liver cirrhosis, anaphylatic shock, disorders of spleen, pernicious anemia, typhoid, paratyphoid, viral infections etc.

- **Leukocytosis:** Increase in WBC count above 11,000 mm3 . It occurs in conditions like infections, allergy, common cold, tuberculosis and glandular fluid.

- **Neutrophilia:** Increase in neutrophils occurs during acute infections, metabolic disorders, chemical poisoning and drugs like lead, mercury, benzene derivatives etc and after acute hemorrhage.

- **Eosinophila:** Increase in eosinophils occurs during allergic conditions, asthma, blood parasitism and scarlet fever.

- **Eosinopenia:** Decrease in eosinophils occurs during small pox, chicken pox and polycythemia.

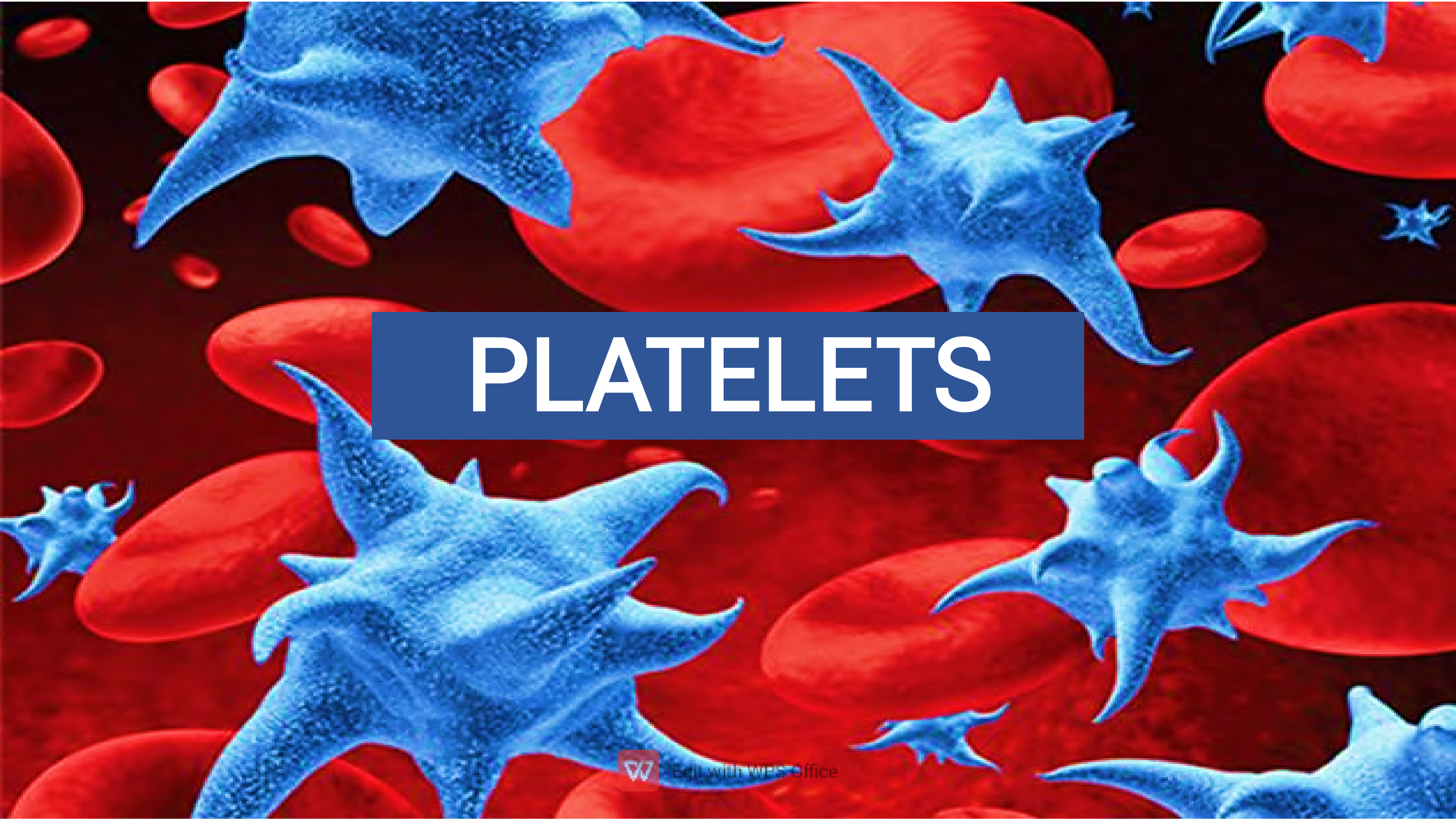
- **Monocytosis:** Increase in monocytes occurs during tuberculosis, syphilis, malaria, kala-azar, glandular fever etc.

- **Lymphocytosis:** Increase in lymphocytes occurs during diphtheria, infectious hepatitis, mumps, mal nutrition, rickets, syphilis, tuberculosis etc.

- **Leukemia:** It is an abnormal and uncontrolled proliferation of leukopoietic tissue. It is always associated with an abnormal increase in blast cells in bone marrow.
- Signs and symptoms- fatigue, recurrent infections, anemia, bone and joint pain, abdominal distress, dyspnea etc.
- There are 2 types,

1. **Myeloid Leukemia:** Where myelocytes are greater in number.
2. **Lymphoid Leukemia:** Increased number of lymphocytes

PLATELETS



- Also called **Thrombocytes.**
- Main role in the **arrest of bleeding.**
- Normal value **2-5 lakhs/cu mm of blood.**
- **Life span 8-11 days.**
- Platelets are small non- nucleated discs which contain mitochondria with various enzymes, contractile proteins like actin, myosin etc.
- They are also **phagocytic and phagocytize antigen – antibody complexes.**

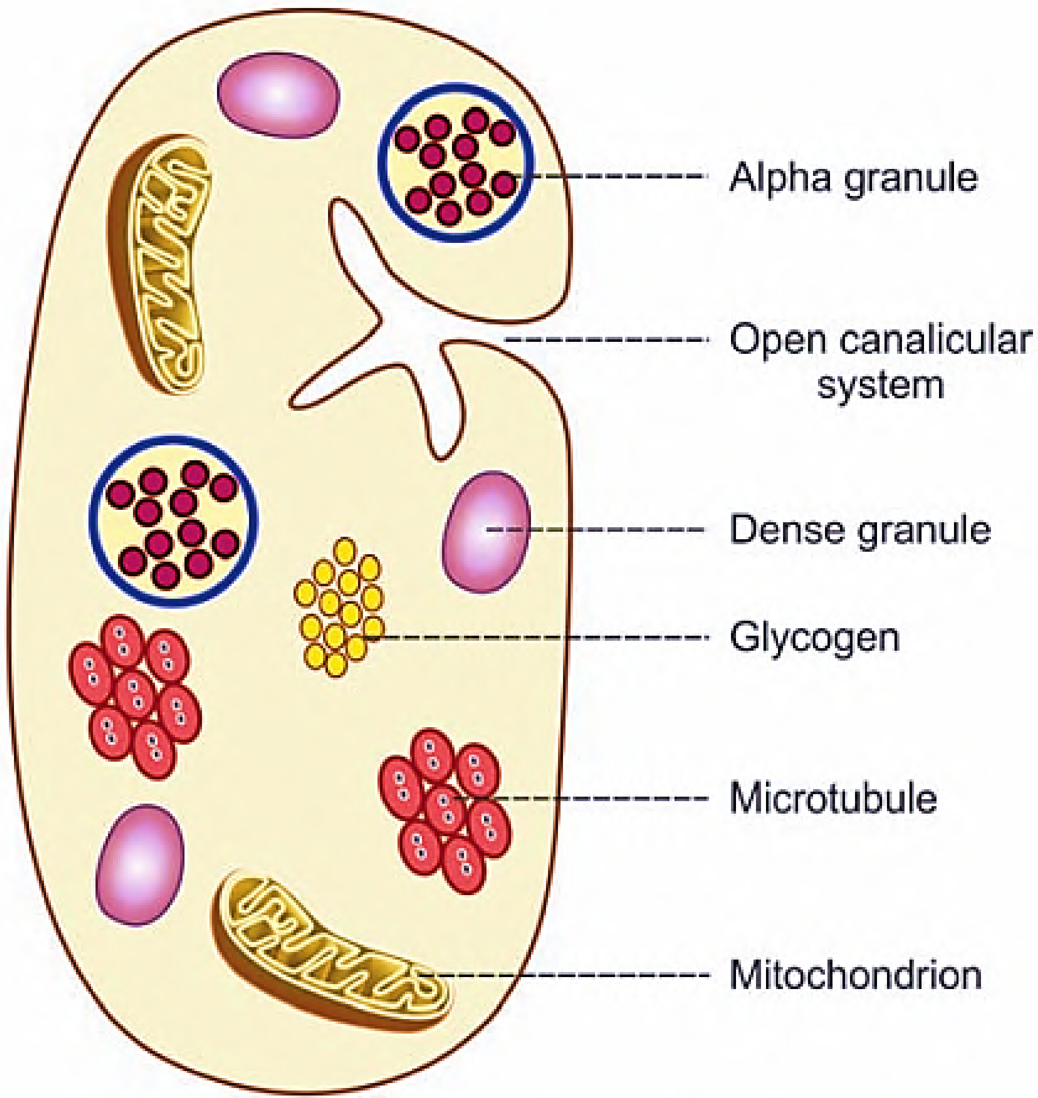
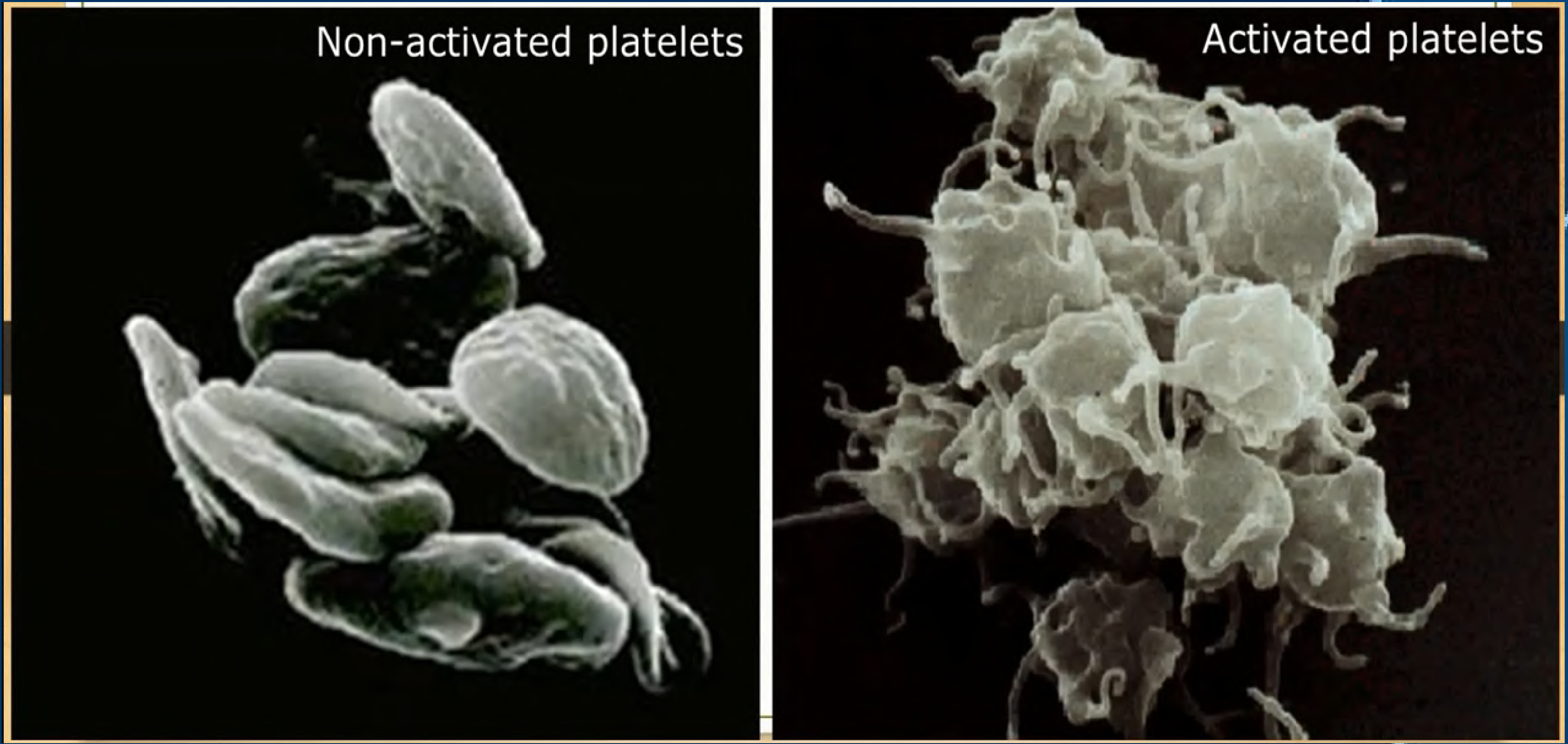


FIGURE 18.1: Platelet under electron microscope

Properties of Platelets:

- **Adhesiveness:** it is the property of sticking to a rough surface.
- **Aggregation:** It is the grouping of platelets.
- **Agglutination:** Clumping together of platelets.


Non-activated platelets
Activated platelets

Functions of Platelets:

1. **Role in blood clotting:** platelets are responsible for the formation of intrinsic prothrombin activator. This substance is responsible for the onset of blood clotting.

2. **Role in prevention of blood loss:** By formation of temporary plug, also platelets seal the damage in blood vessels.

3. **Role in clot retraction:** In the blood clot, the blood cells including platelets are entrapped in between the fibrin threads. The cytoplasm of platelets contains the contractile proteins, which are responsible for clot retraction (More compact structure).
4. **Role in repair of ruptured blood vessels:** Platelet Derived Growth Factor (PDGF) is useful for the repair of the endothelium and other structures of the ruptured blood vessels.
5. **Role in defense mechanism:** By the property of agglutination, platelets encircle the foreign bodies and destroy them by phagocytosis

Physiological variations:

1. **Age:** Less in infants and reaches normal level at 3rd month after birth.

2. **Gender:** No difference between males and females. In females it is reduced during menstruation.

3. **High altitude:** Platelet count increases.

4. **After meals:** Platelet count increases.

Pathological variations:

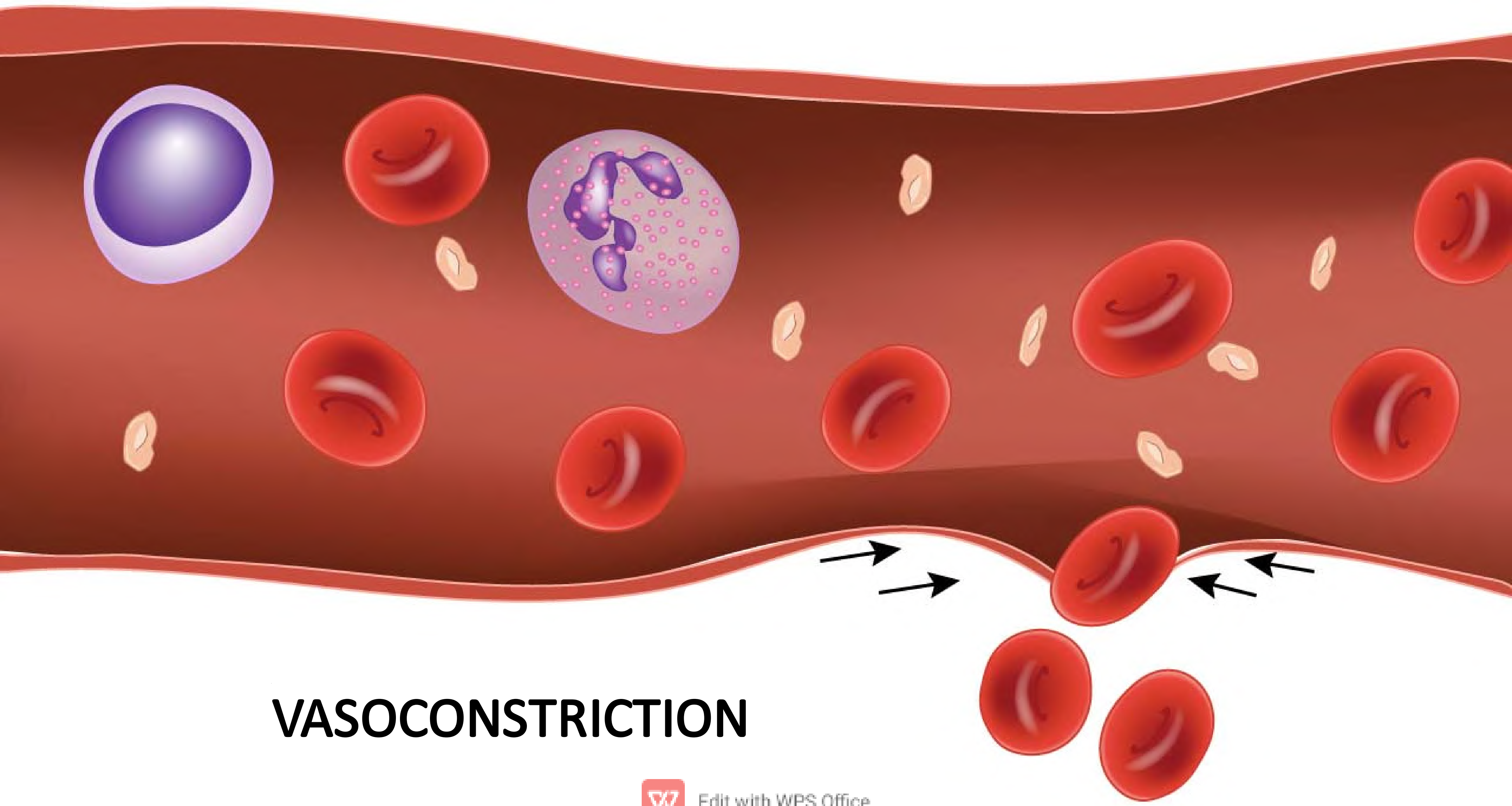
- Thrombocytopenia: Decrease in platelet count.
- Thrombocytosis: Increase in platelet count.
- Thrmbocythemia: Persistent and abnormal increase in platelet count.
- Glanzmann thrombasthenia: Inherited hemorrhagic disorder caused by structural or functional abnormality of platelets.


HEMOSTASIS

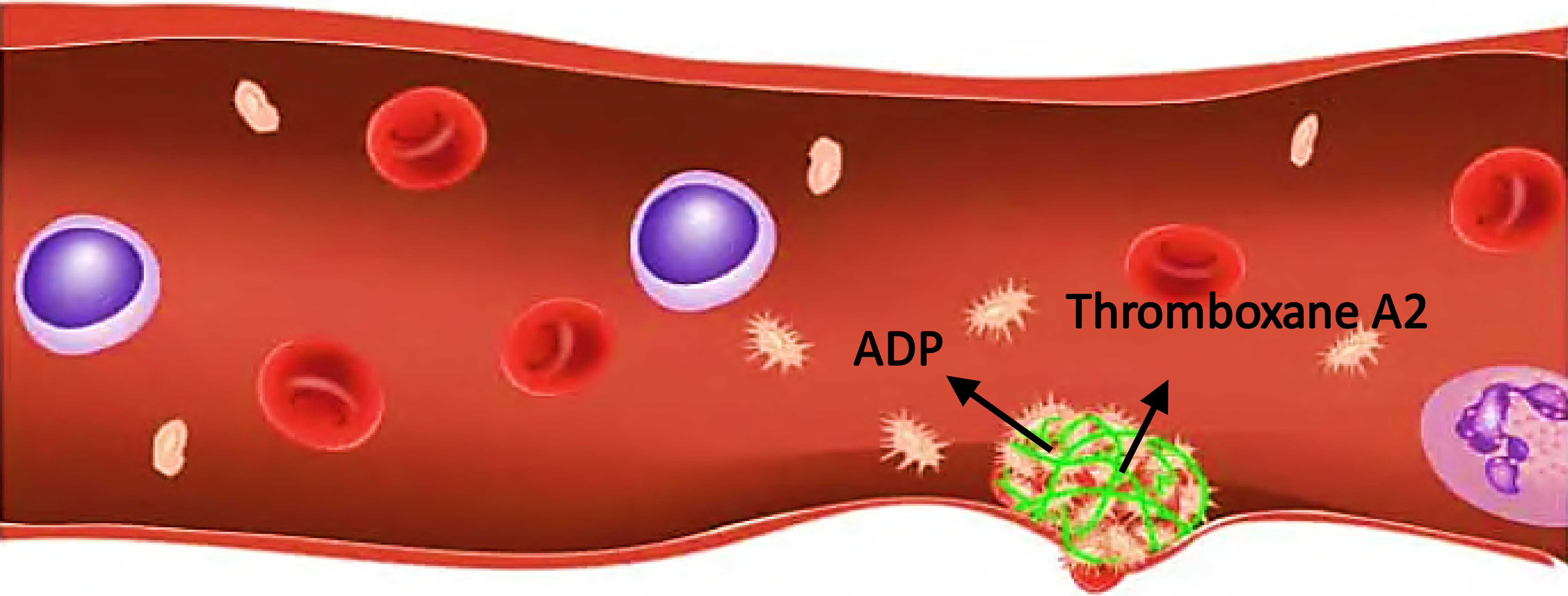
- Arrest or stoppage of bleeding by physiological process.

- Involves 3 steps or stages:

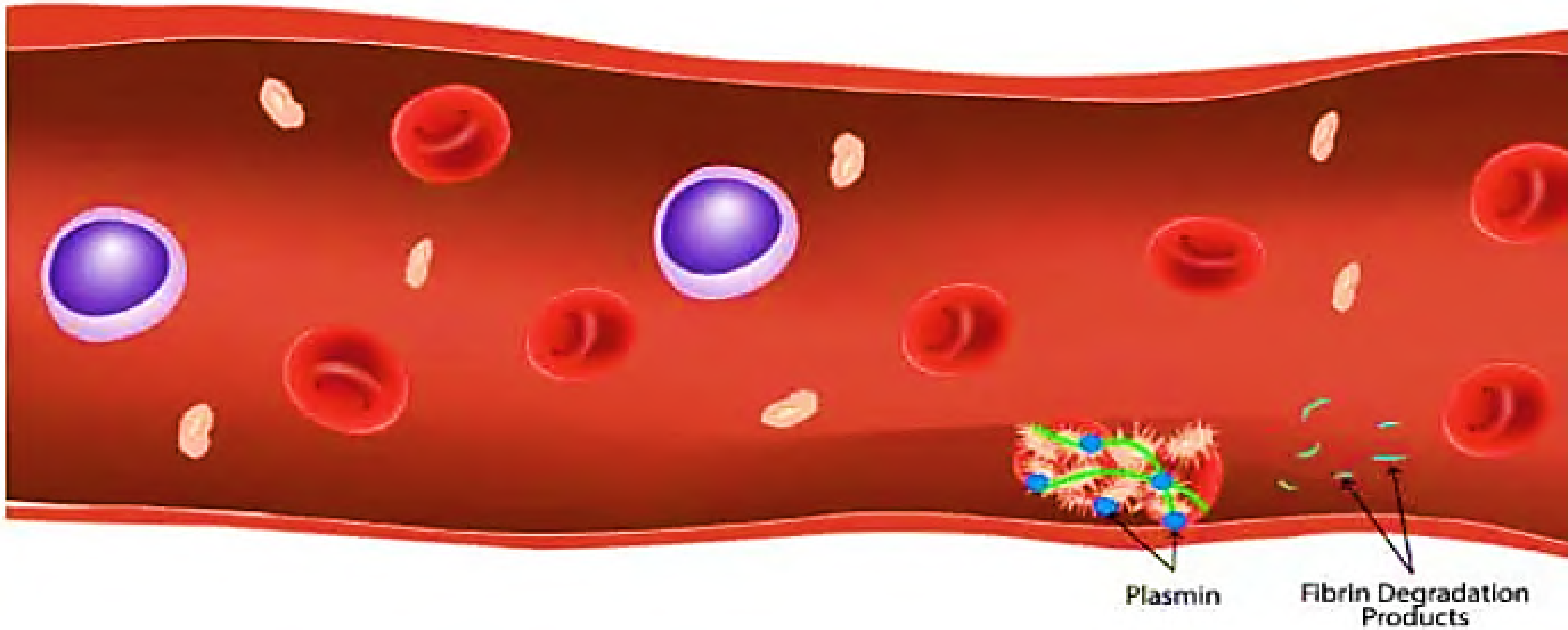
1. Vasoconstriction
2. Platelet plug formation
3. Coagulation of blood/blood clotting



VASOCONSTRICTION



PLATELET PLUG FORMATION



COAGULATION OF BLOOD

Vasoconstriction:

- **Constriction of blood vessels** just before the area of injury/damage.
- At the point of injury, **platelets adhere** on the exposed collagen fibers (rough surface).
- Activated platelets release vasoconstrictors like **serotonin (5-HT)** and **nor- epinephrine** that leads to vasoconstriction.
- This helps in temporary prevention of blood loss.

Platelet plug formation:

- Also known as **temporary hemostatic plug.**
- Activated platelets adhere, aggregate and agglutinate at the point of injury to **form platelet plug.**
- **Temporarily seals** the injured area thereby decreasing blood loss.
- Helps in the **activation of clotting factors** leading to blood coagulation.

Blood coagulation:

- Also called **Permanent hemostatic plug.**

- Process by which blood loses its fluid form and gets converted into a Jelly like substance at the point of injury to prevent further blood loss.

- Occurs by the activation of clotting factors in blood.

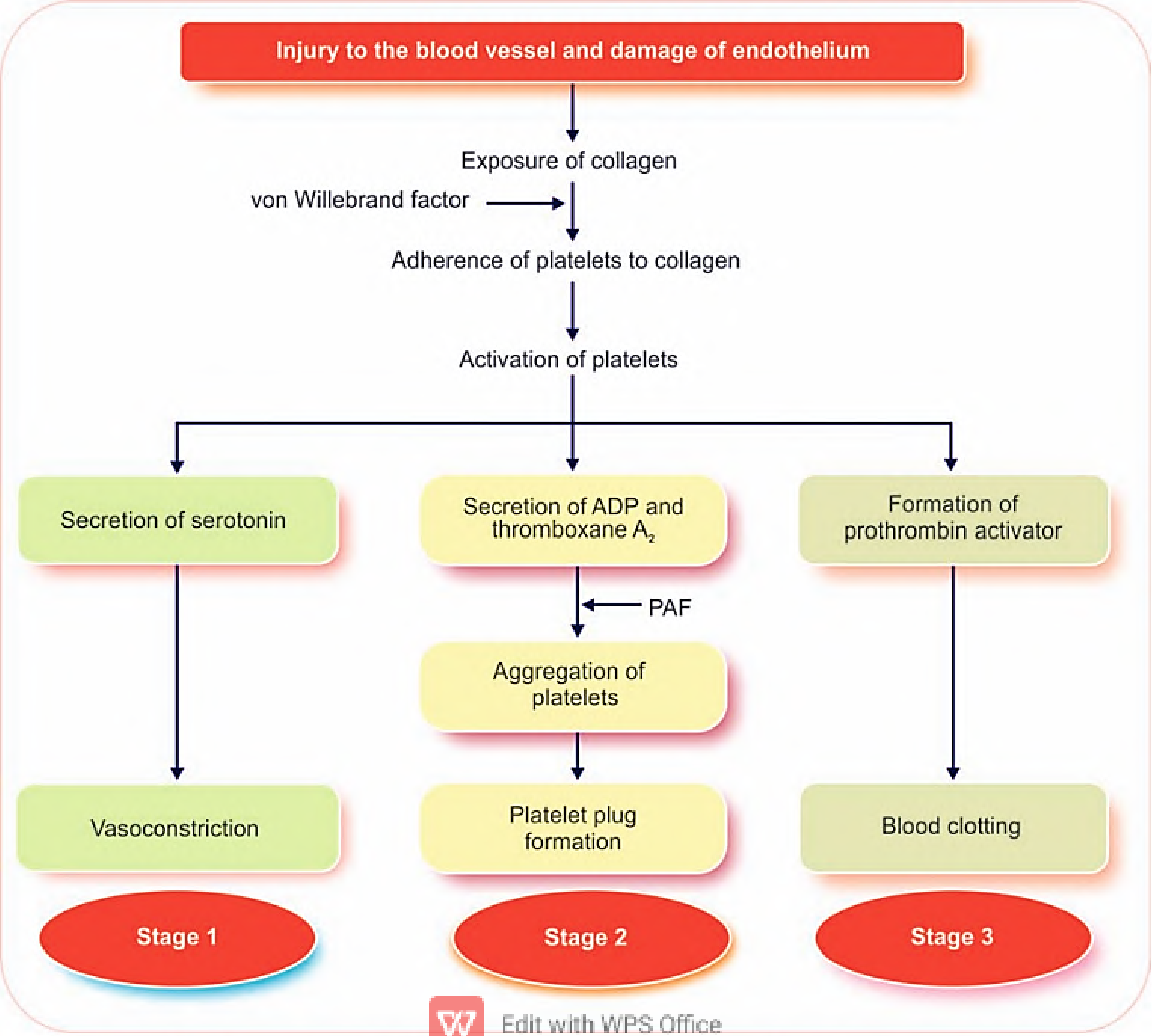
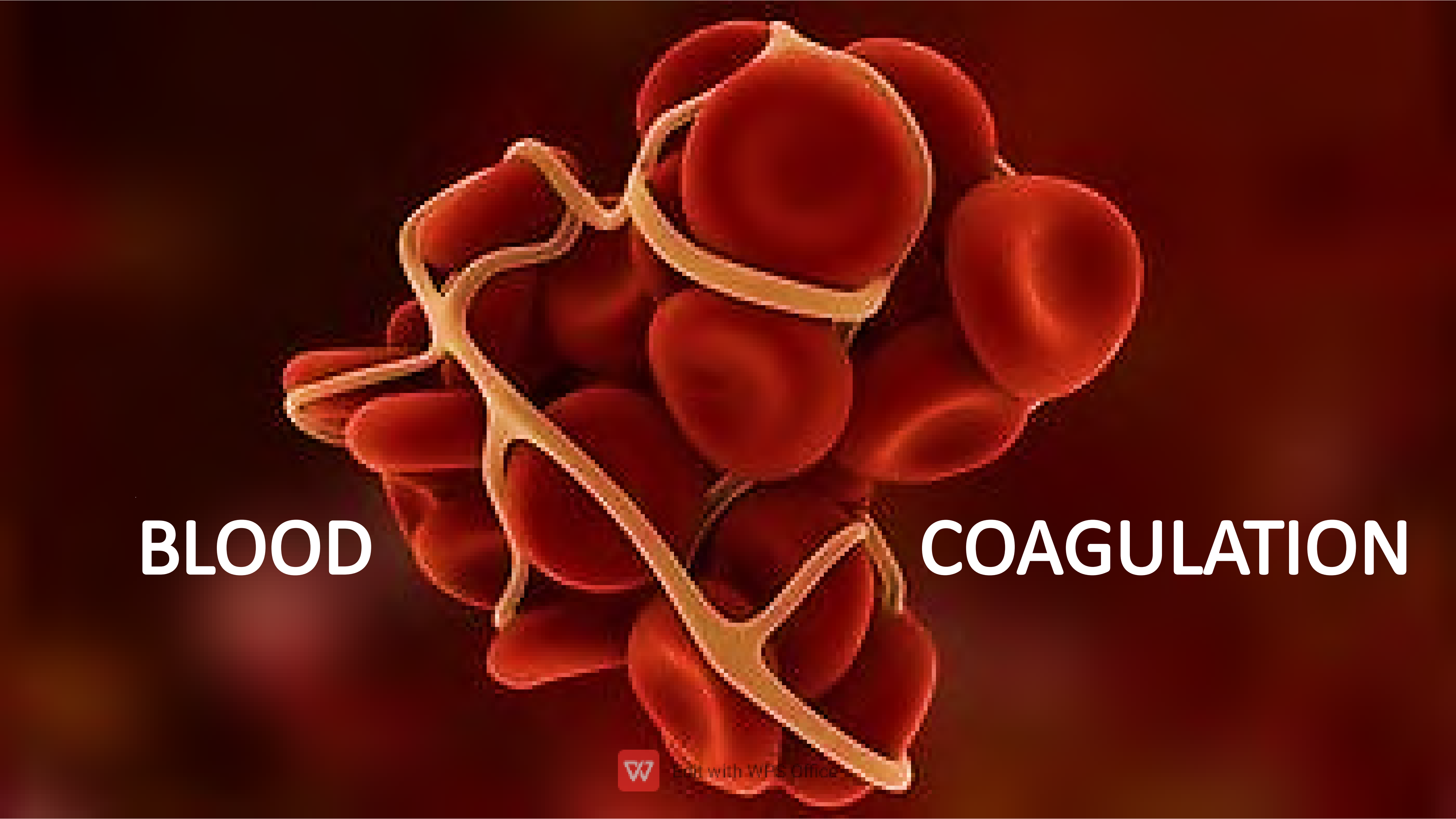


FIGURE 19.1: States of hemostasis. ADP = Adenosine diphosphate; PAF = Platelet-activating factor.


BLOOD
COAGULATION

- Coagulation or clotting is **defined as the process in which liquid blood is converted into jelly like mass.**

- Coagulation of blood occurs through **a series of reactions** due to the activation of a group of substances.

Clotting factors:

- Factor I : Fibrinogen
- Factor II: Prothrombin
- Factor III: Thrompoplastin
- Factor IV : Calcium
- Factor V : Labile factor
- Factor VI : Unknown factor
- Factor VII : Stable factor
- Factor VIII : Antihemophilic factor A
- Factor IX : Antihemophilic factor B /Christmas factor
- Factor X : Stuart Prower factor
- Factor XI : Antihemophilic factor C or Plasma thromboplastin
- Factor XII : Hageman factor
- Factor XIII : Fibrin stabilizing factor

Clotting factors:

- Factor I : Fibrinogen
- Factor II: Prothrombin
- Factor III: Thrompoplastin
- Factor IV : Calcium
- Factor V : Labile factor
- Factor VI : Unknown factor
- Factor VII : Stable factor
- Factor VIII : Antihemophilic factor A
- Factor IX : Antihemophilic factor B /Christmas factor
- Factor X : Stuart Prower factor
- Factor XI : Antihemophilic factor C or Plasma thromboplastin
- Factor XII : Hageman factor
- Factor XIII : Fibrin stabilizing factor

Fiber Provide ചെയ്യലാണ് Thampurante joli. Thampuranu Ca ഉണ്ടാക്കുന്ന Lab und. അതിന് പേരില്ല (unknown). എങ്കിലും അത് stable ആയിട്ട് പോണുണ്ട് .

തമ്പുരാന് 8 anti - മാർ ഉണ്ട്. 8 പേരും ഗർഭിണികൾ ആണ്..

9 - ആമത്തേ മാസം അവർ christmas 🎄 ആഘോഷിച്ചു.

10 - ആമത്തെ മാസം Stuart കുട്ടി ജനിച്ചു.

തമ്പുരാന്റെ പേരും കൂട്ടി ഒരു പേരിട്ടു വിളിച്ചു (plasma Thromboplastin).

ജനങ്ങൾ അവനെ "HAGEMAN" എന്ന് വിളിച്ച് പ്രശംസിച്ചു.

അങ്ങനെ അവൻ fiber കമ്പനി stable ആക്കി നടത്തി പോവുന്നു.

Stages of blood clotting:

1. Formation of prothrombin activator in 2 pathways:

a. **Extrinsic pathway** (from injured tissue)

b. **Intrinsic pathway** (within the blood itself/damage to blood vessels)

2. Conversion of prothrombin to thrombin.

3. Conversion of fibrinogen to fibrin.

Intrinsic pathway

Intrinsic pathway

Endothelial damage + collagen exposure

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein ↓

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein ⬇

XII

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein

XII $\longrightarrow$

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein

XII → XIIa

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein

XII → XIIa

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein ⬇

XII → XIIa

XI

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein

$$\begin{array}{ccc} XII & \longrightarrow & XIIa \\ & \swarrow & \\ XI & \longrightarrow & \end{array}$$

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein ↓

XII → XIIa

XI → XIa

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein ↓

XII → XIIa

XI → XIa

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein

XII → XIIa

XI → XIa

IX

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein

XII → XIIa

XI → XIa

IX →

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein ↓

XII → XIIa

XI → XIa

IX → IXa

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein ⬇

$$\begin{tikzcd}
\mathrm{XII} \arrow[r] & \mathrm{XIIa} \arrow[dl] \\
\mathrm{XI} \arrow[r] & \mathrm{XIa} \arrow[dl] \\
\mathrm{IX} \arrow[r] & \mathrm{IXa} \arrow[dl] \\
 &
\end{tikzcd}$$

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein ↓

XII → XIIa

XI → XIa

IX → IXa

VIII

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein ↓

XII → XIIa

XI → XIa

IX → IXa

VIII →

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein

XII → XIIa

XI → XIa

IX → IXa

VIII → VIIIa

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein

XII → XIIa

XI → XIa

IX → IXa

VIII → VIIIa

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein

XII → XIIa

XI → XIa

IX → IXa

VIII → VIIIa

Va, Ca2+

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein ⇓

XII → XIIa

XI → XIa

IX → IXa

VIII → VIIIa

Va, Ca2+

X

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein ⇩

XII → XIIa

XI → XIa

IX → IXa

VIII → VIIIa

Va, Ca2+

X →

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein

XII → XIIa

XI → XIa

IX → IXa

VIII → VIIIa

Va, Ca2+

X → Xa

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein

XII → XIIa

XI → XIa

IX → IXa

VIII → VIIIa

Va, Ca^{2+}

X → Xa

Intrinsic pathway

Endothelial damage + collagen exposure → platelets

HMW kinogen, kallikrein

XII → XIIa

XI → XIa

IX → IXa

VIII → VIIIa

Va, Ca2+

X → Xa

Intrinsic pathway

Endothelial damage + collagen exposure → platelets

HMW kinogen, kallikrein

XII → XIIa

XI → XIa

IX → IXa

VIII → VIIIa

Va, Ca2+

X → Xa

Intrinsic pathway

Endothelial damage + collagen exposure

HMW kinogen, kallikrein

XII → XIIa

XI → XIa

IX → IXa

VIII → VIIIa

Va, Ca^{2+}

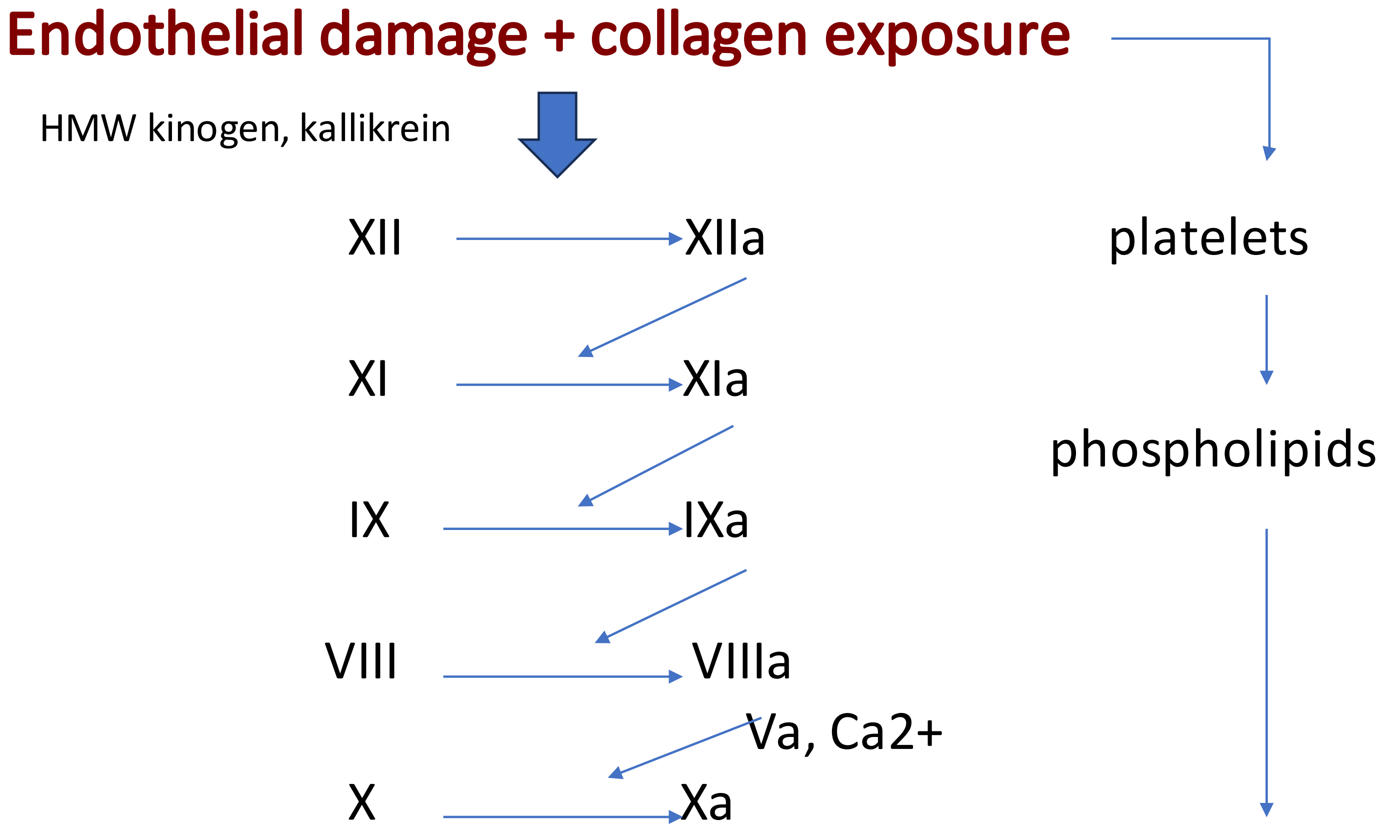
X → Xa

platelets

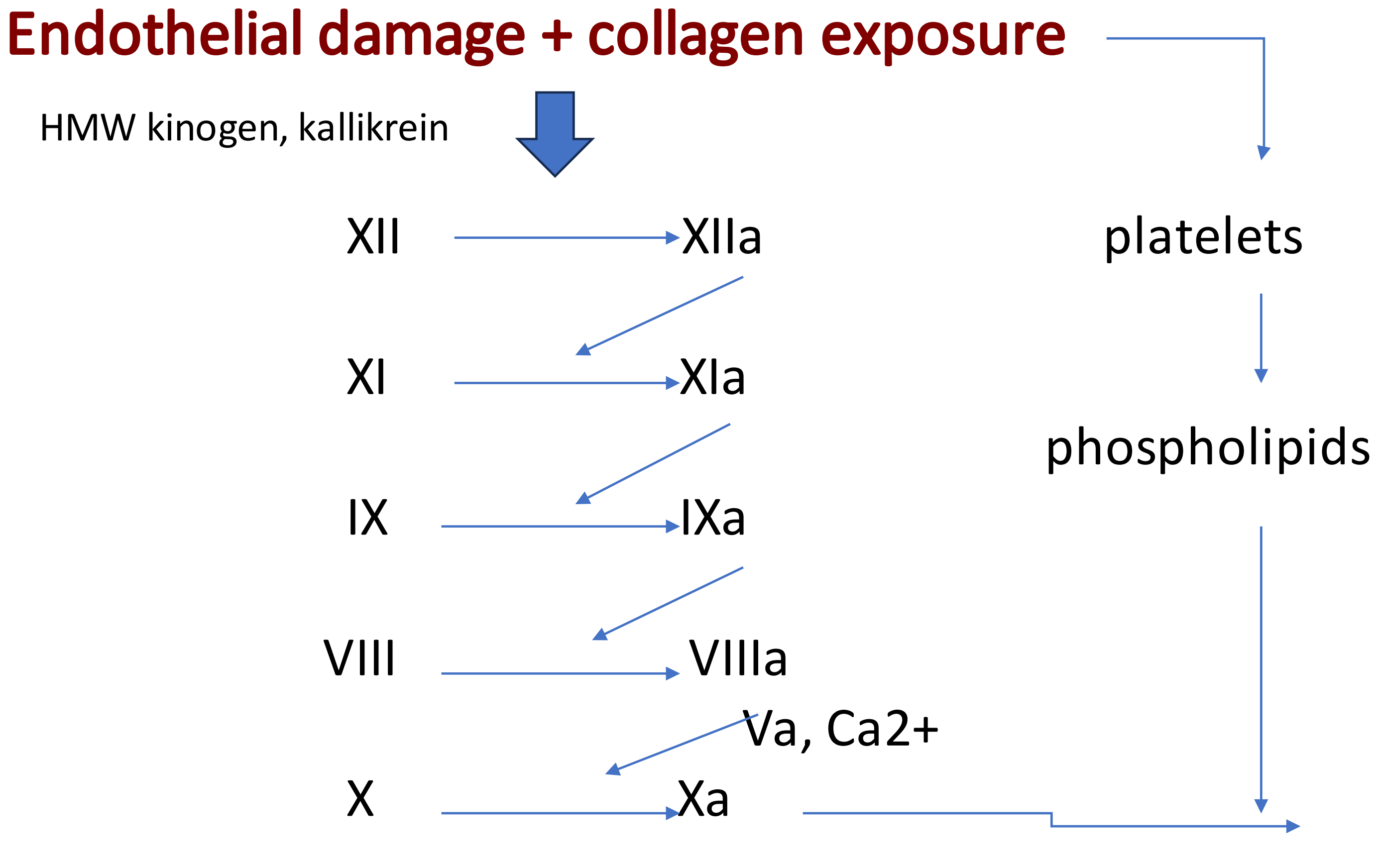
↓

phospholipids

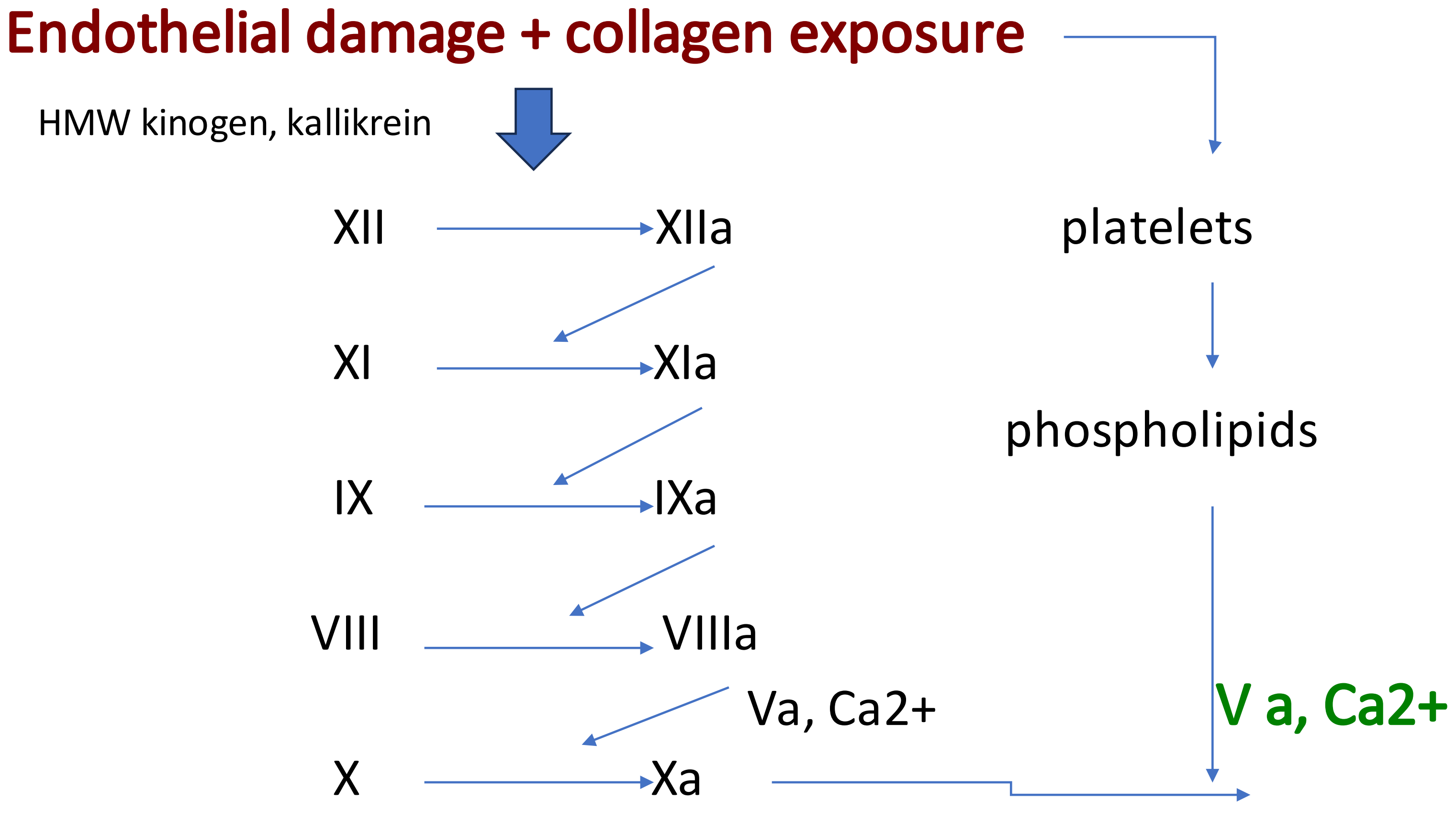
Intrinsic pathway



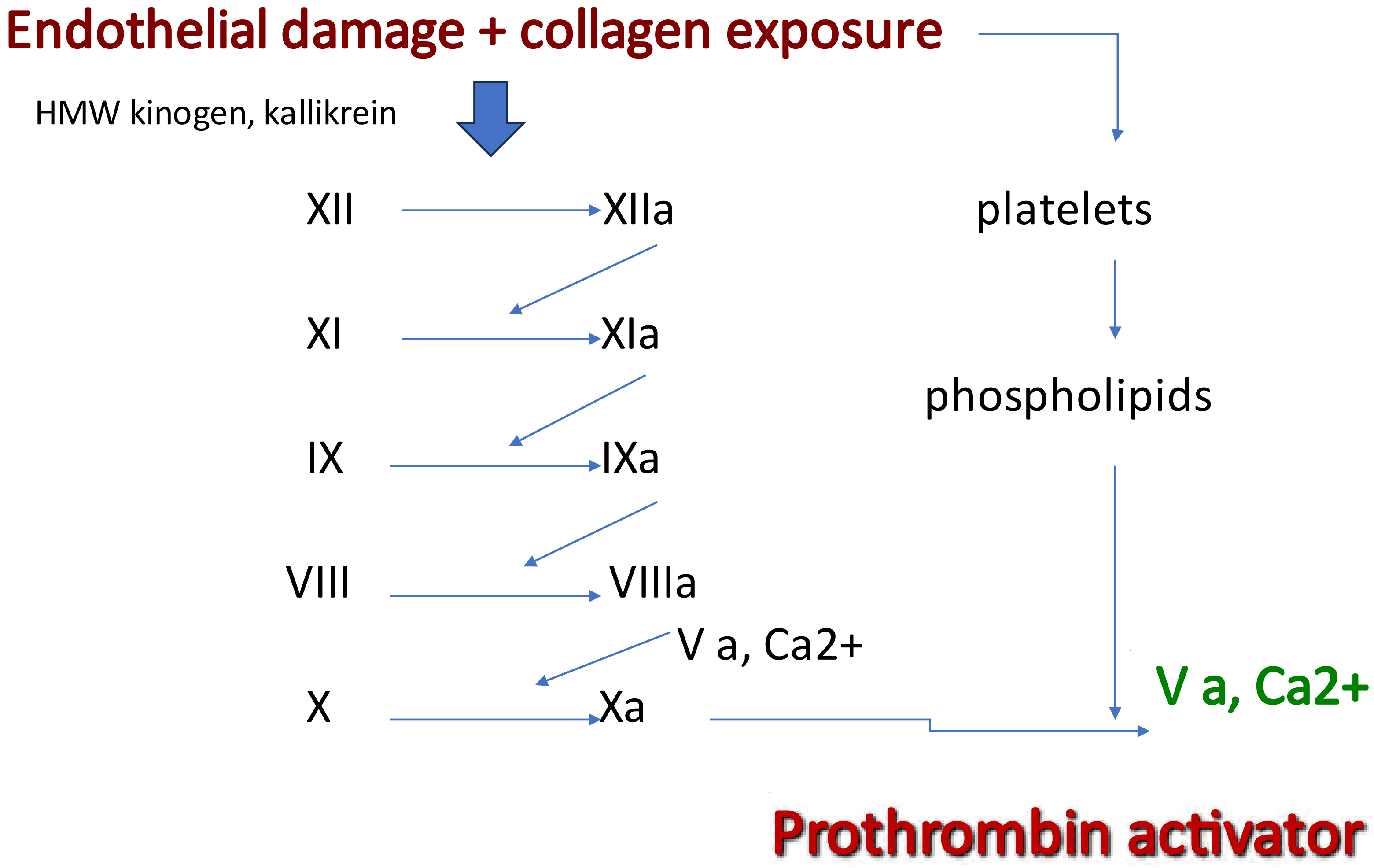
Intrinsic pathway



Intrinsic pathway



Intrinsic pathway



Extrinsic pathway

Extrinsic pathway

Tissue trauma + tissue thromboplastin (factor III)

Extrinsic pathway

Tissue trauma + tissue thromboplastin (factor III)

↓

Glycoprotein & phospholipid

Extrinsic pathway

Tissue trauma + tissue thromboplastin (factor III)

↓

Glycoprotein phospholipid

↓

Extrinsic pathway

Tissue trauma + tissue thromboplastin (factor III)

↓

Glycoprotein phospholipid

↓

VII

Extrinsic pathway

Tissue trauma + tissue thromboplastin (factor III)

↓

Glycoprotein phospholipid

↓

VII ⟶

Extrinsic pathway

Tissue trauma + tissue thromboplastin (factor III)

$\downarrow$

Glycoprotein phospholipid

$\downarrow$

VII $\longrightarrow$ VIIa

Extrinsic pathway

Tissue trauma + tissue thromboplastin (factor III)

↓

Glycoprotein phospholipid

↓

VII → VIIa

↙

Extrinsic pathway

Tissue trauma + tissue thromboplastin (factor III)

$\downarrow$

Glycoprotein phospholipid

$\downarrow$

VII $\longrightarrow$ VIIa

X

Extrinsic pathway

Tissue trauma + tissue thromboplastin (factor III)

↓

Glycoprotein phospholipid

↓

VII → VIIa

↙

X →

Extrinsic pathway

Tissue trauma + tissue thromboplastin (factor III)

↓

Glycoprotein phospholipid

↓

VII → VIIa

X → Xa

Extrinsic pathway

Tissue trauma + tissue thromboplastin (factor III)

↓

Glycoprotein phospholipid

↓

VII → VIIa

↙

X → Xa

Calcium, Va

→

Extrinsic pathway

Tissue trauma + tissue thromboplastin (factor III)

↓

Glycoprotein phospholipid

↓

VII → VIIa

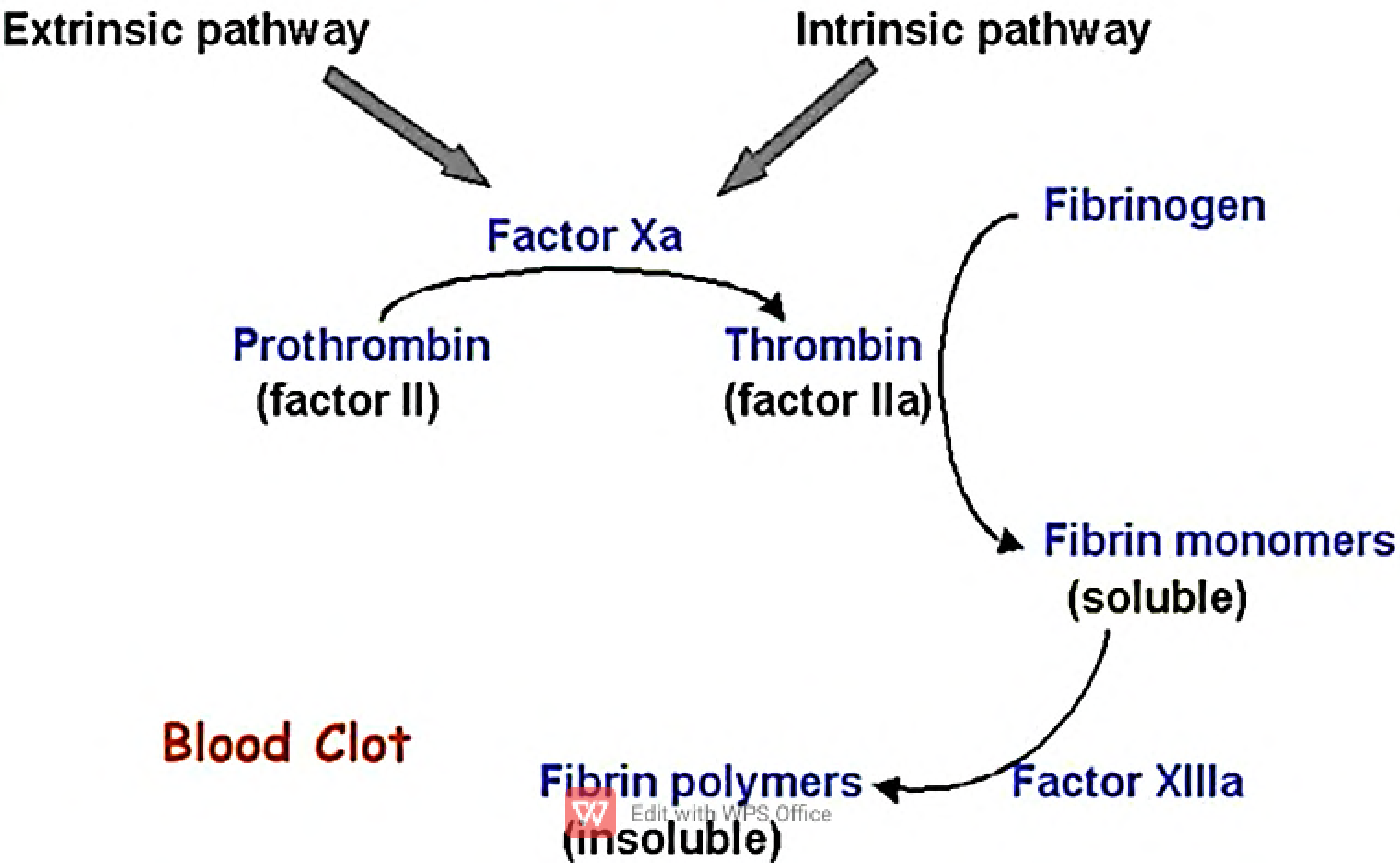
↙

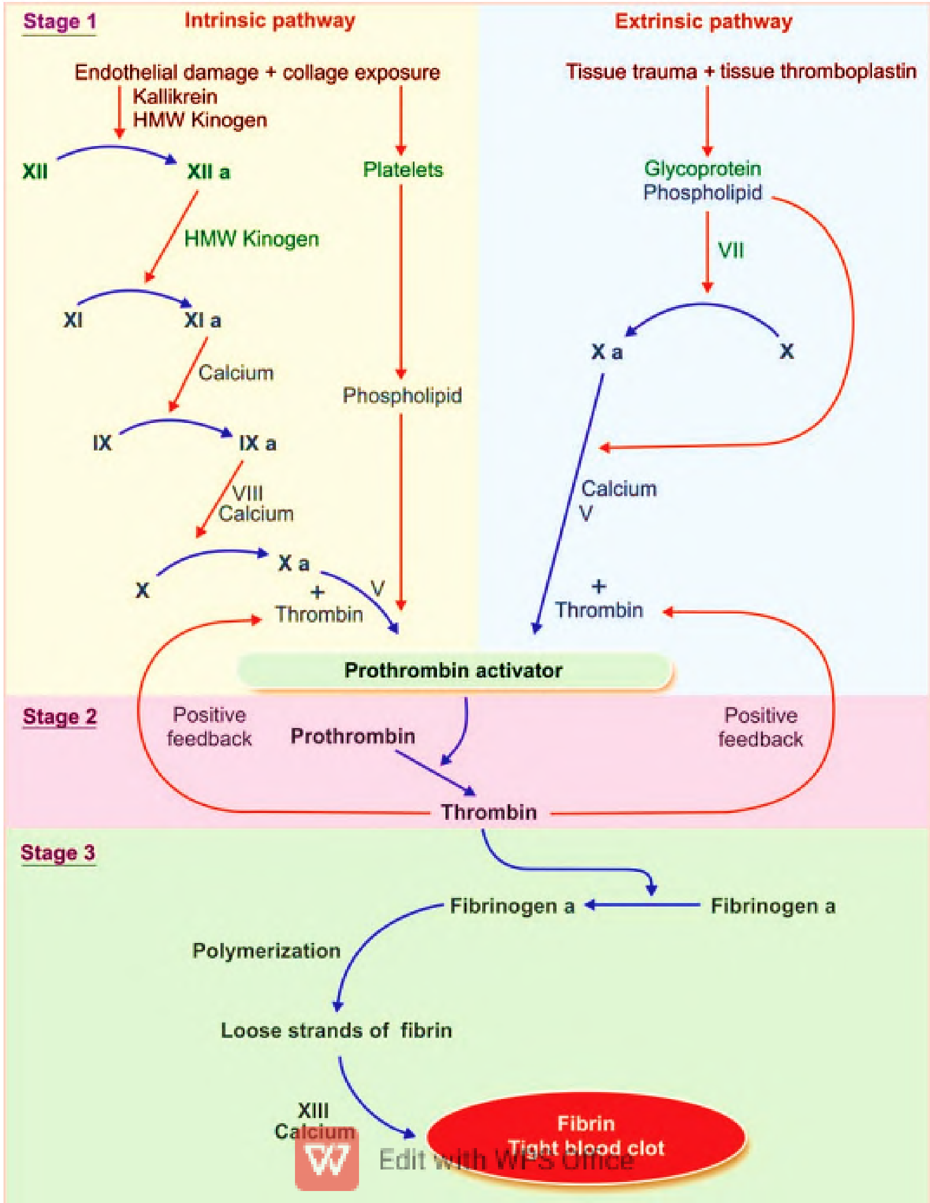
X → Xa

Calcium, Va

→ **Prothrombin activator**

Fibrin Formation




Stage 1
Intrinsic pathway
Extrinsic pathway
Endothelial damage + collage exposure
Kallikrein
HMW Kinogen
Tissue trauma + tissue thromboplastin
XII
XII a
Platelets
Glycoprotein
Phospholipid
HMW Kinogen
VII
XI
XI a
X a
X
Calcium
Phospholipid
IX
IX a
Calcium
V
VIII
Calcium
X
X a
+
Thrombin
V
+
Thrombin
Prothrombin activator
Stage 2
Positive
feedback
Prothrombin
Positive
feedback
Thrombin
Stage 3
Fibrinogen a
Fibrinogen a
Polymerization
Loose strands of fibrin
XIII
Calcium
Fibrin
Tight blood clot

Tests of coagulation:

- **Bleeding time:** Time interval between onset of bleeding till the stoppage of bleeding. Normal value is 2 - 4 minutes.

- **Clotting time:** Time interval between the onset of bleeding till the formation of fibrin thread. Normal value is 3 – 8 minutes.

- **Prothrombin time:** it measures the efficiency of extrinsic pathway and the common pathway of coagulation. Normal value is 16 seconds.

Bleeding Disorders:

1. Haemophilia
2. Purpura
3. Von Willebrand disease
4. Thrombosis

Haemophilia:

- **Sex linked blood disorder** caused due to the deficiency of clotting factors.
- 2 types: -
  a. **Hemophilia A/ Classic hemophilia-** most common type of hemophilia caused due to the deficiency of clotting factor VIII (anti- hemophilic factor)
  b. **Hemophilia B/ Christmas disease-** rare type of hemophilia caused due to the deficiency of clotting factor IX (Christmas factor).

Purpura:

- Also called **blood spots or skin hemorrhages** refers to purple-colored spots underneath the skin.

- Occurs secondary to **decreased platelet count**, coagulation disorders etc.

- Decreased platelet count **results in poor clot retraction** and **poor constriction of injured blood vessels** leading to pooling of blood under the skin.

Von Willebrand disease

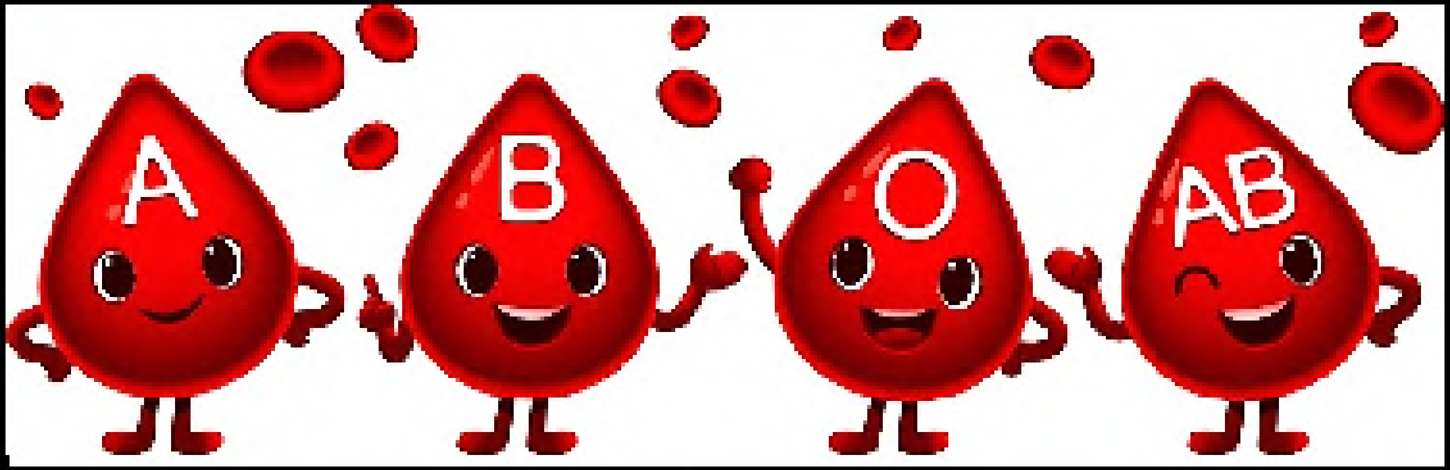
- It is an extremely rare disorder caused due to the **deficiency or absence of von- Willebrand factor** in the platelets.

- Platelet adhesion on to rough surfaces will be affected leading to delayed clotting.

Thrombosis:

- Refers to coagulation of blood inside the blood vessels.
- Causes of Thrombosis

1. Injury to blood vessel
2. Roughened endothelial lining
3. Sluggishness of blood flow
4. Agglutination of RBCs
5. Toxic thrombosis
6. Congenital absence of protein C



Blood Groups

- Discovery – Karl Landsteiner in 1901

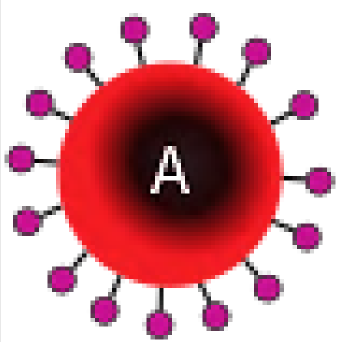
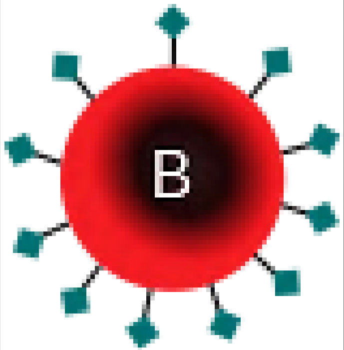
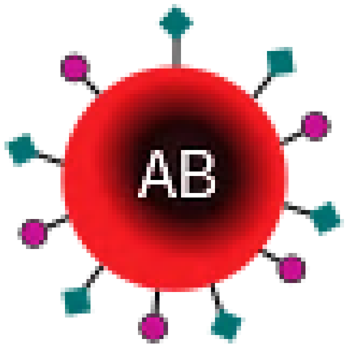
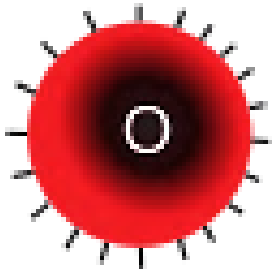
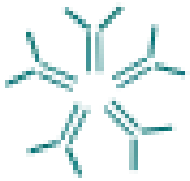
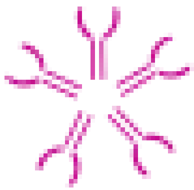
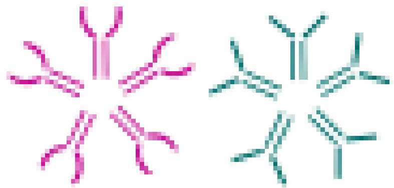
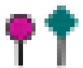
- Based on the presence of blood group **antigens** on the surface of RBCs

- Two types of blood group:
  1. **ABO system**
  2. **Rh system**

ABO BLOOD GROUPS:

- Determination of ABO blood groups depends upon the immunological reaction between antigen and antibody.

ABO SYSTEM

	Group A	Group B	Group AB	Group O
Red blood cell type				
Antibodies in plasma	 Anti-B	 Anti-A	None	 Anti-A and Anti-B
Antigens in red blood cell	 A antigen	 B antigen	 A and B antigens	None

- **Landsteiner law** states that:

1. If a particular agglutinogen (antigen) is present in the RBCs, corresponding agglutinin (antibody) must be absent in the serum.

2. If a particular agglutinogen is absent in the RBCs, the corresponding agglutinin must be present in the serum

Rh FACTOR

- Rh factor is an antigen present in RBC.
- This antigen was discovered by Landsteiner and Wiener.
- It was first discovered in Rhesus monkey and hence the name ‘Rh factor’.
- There are many Rh antigens but only the **D antigen** is more antigenic in human.
- D antigen has no antibody naturally
- If an Rh+ve blood is transfused into Rh -ve blood, there forms anti- D is formed in that person.

Antiserum A -
contains antibody A

Antiserum B -
contains antibody B

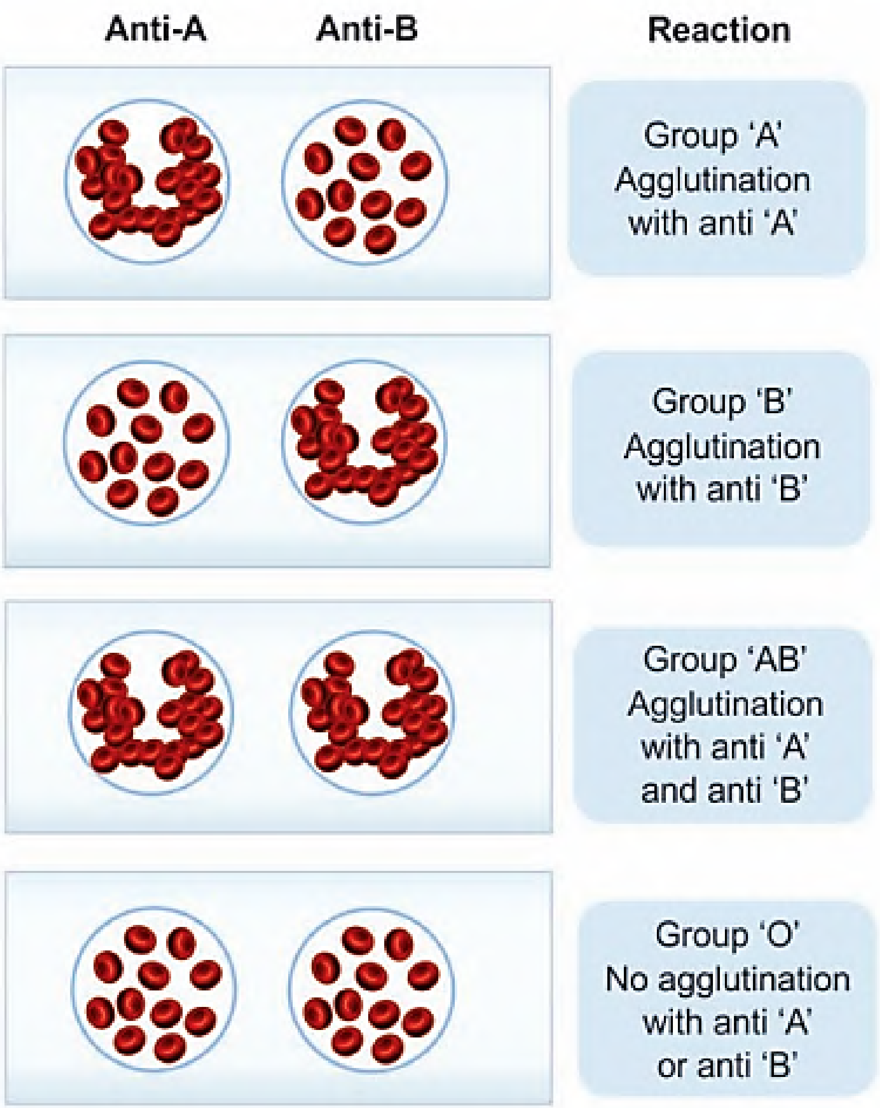
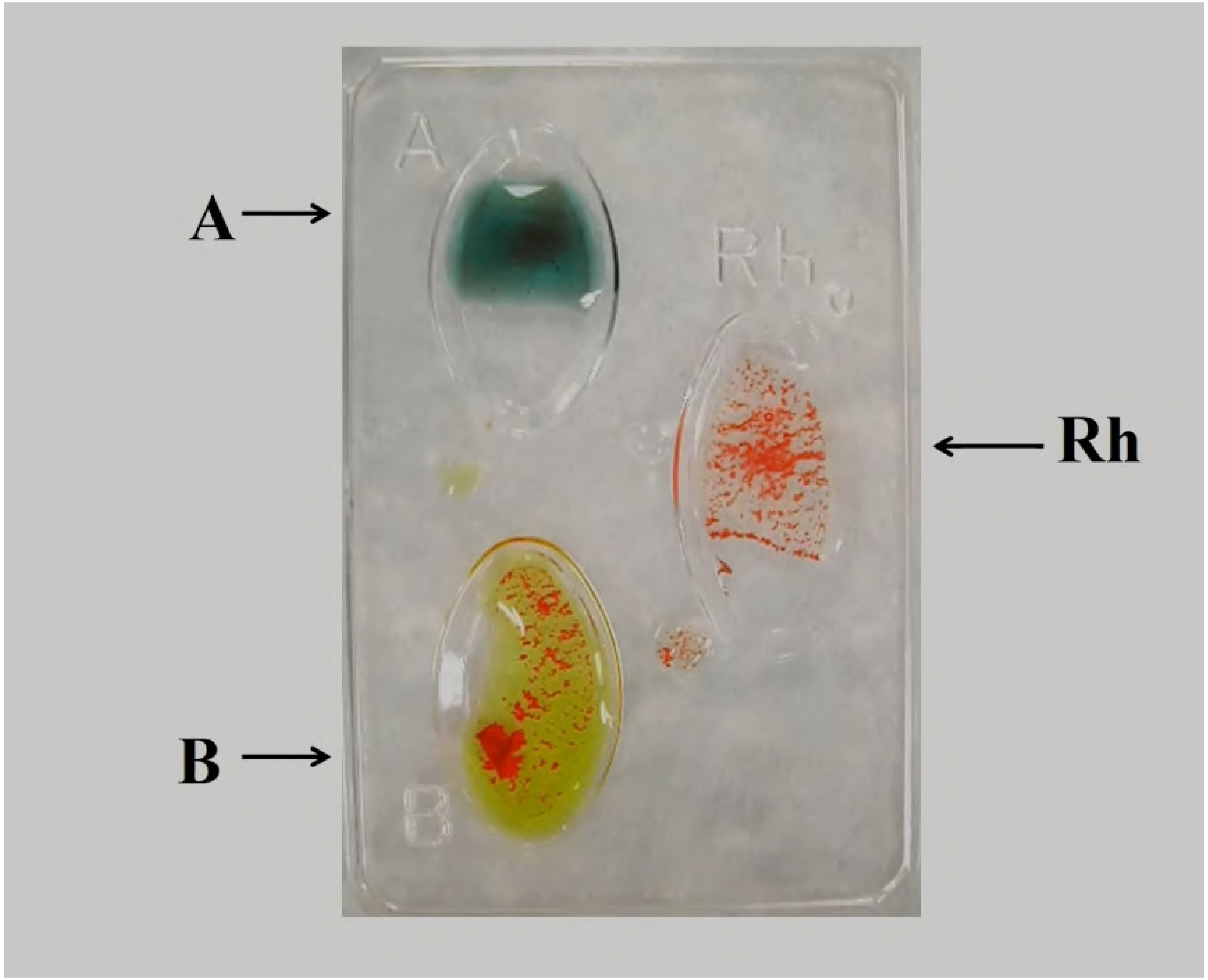
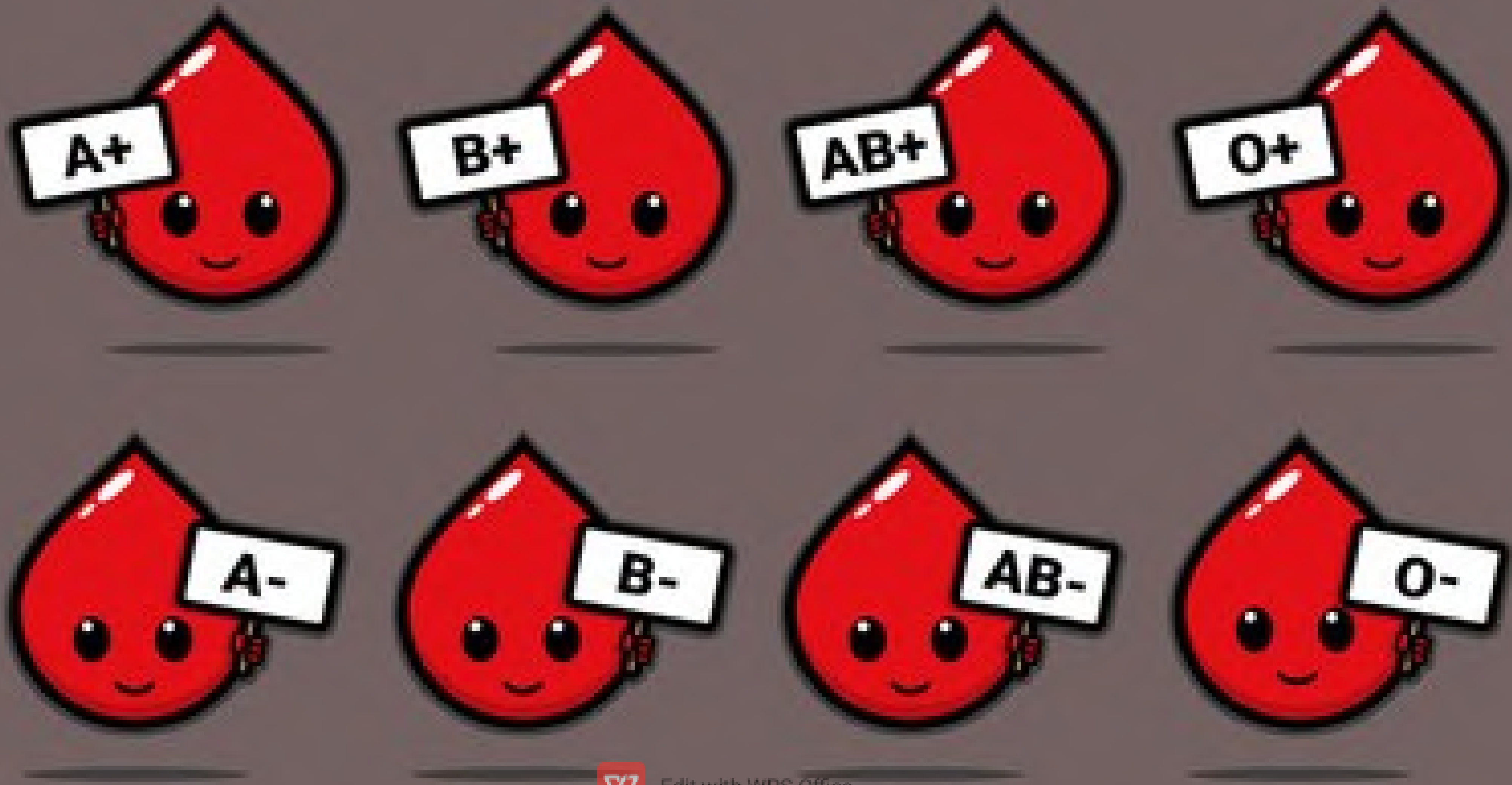


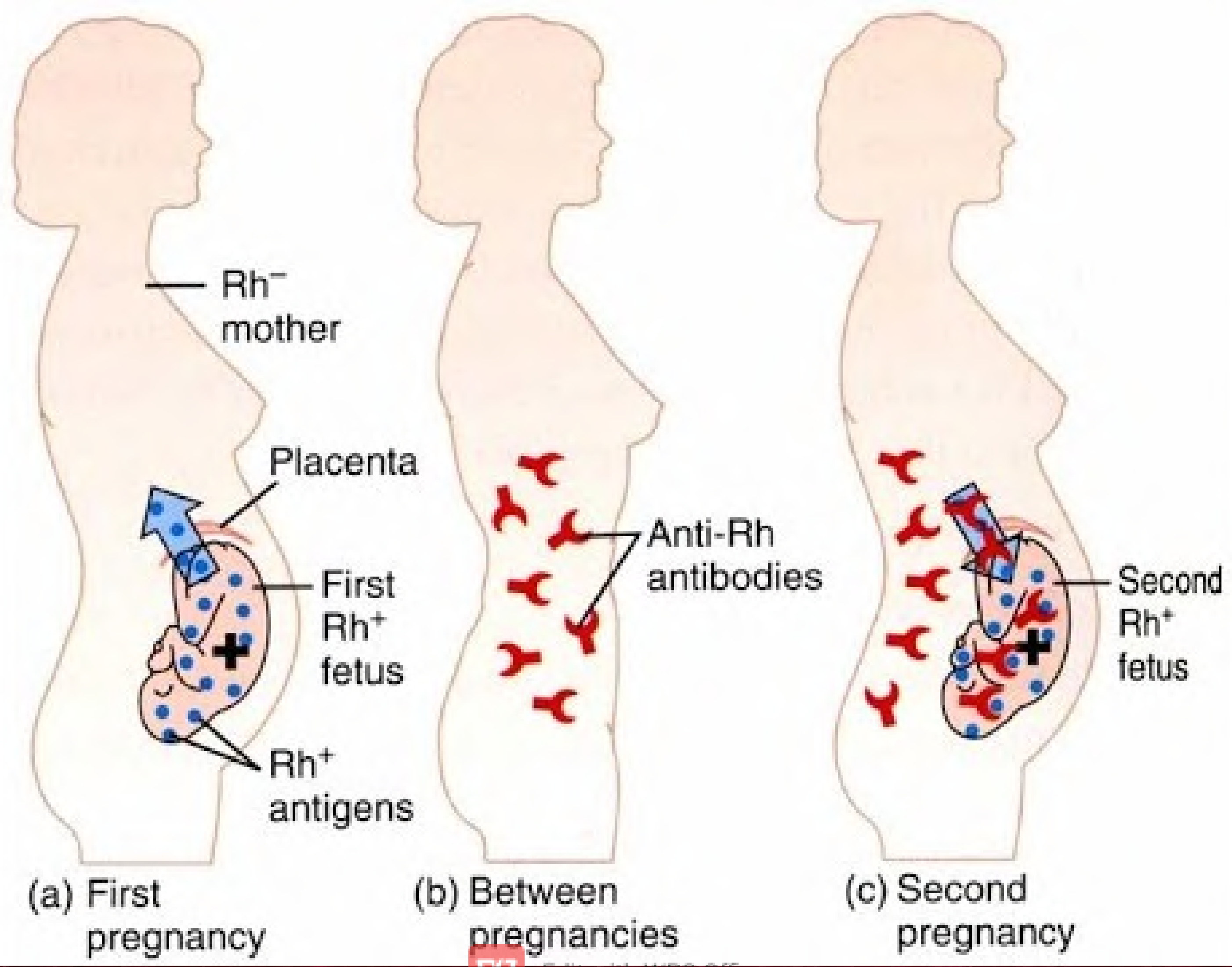
FIGURE 21.1: Determination of blood group


A
Rh
B
A
Rh
B


A+
B+
AB+
O+
A-
B-
AB-
O-

Hemolytic disease of fetus and newborn – Erythroblastosis fetalis

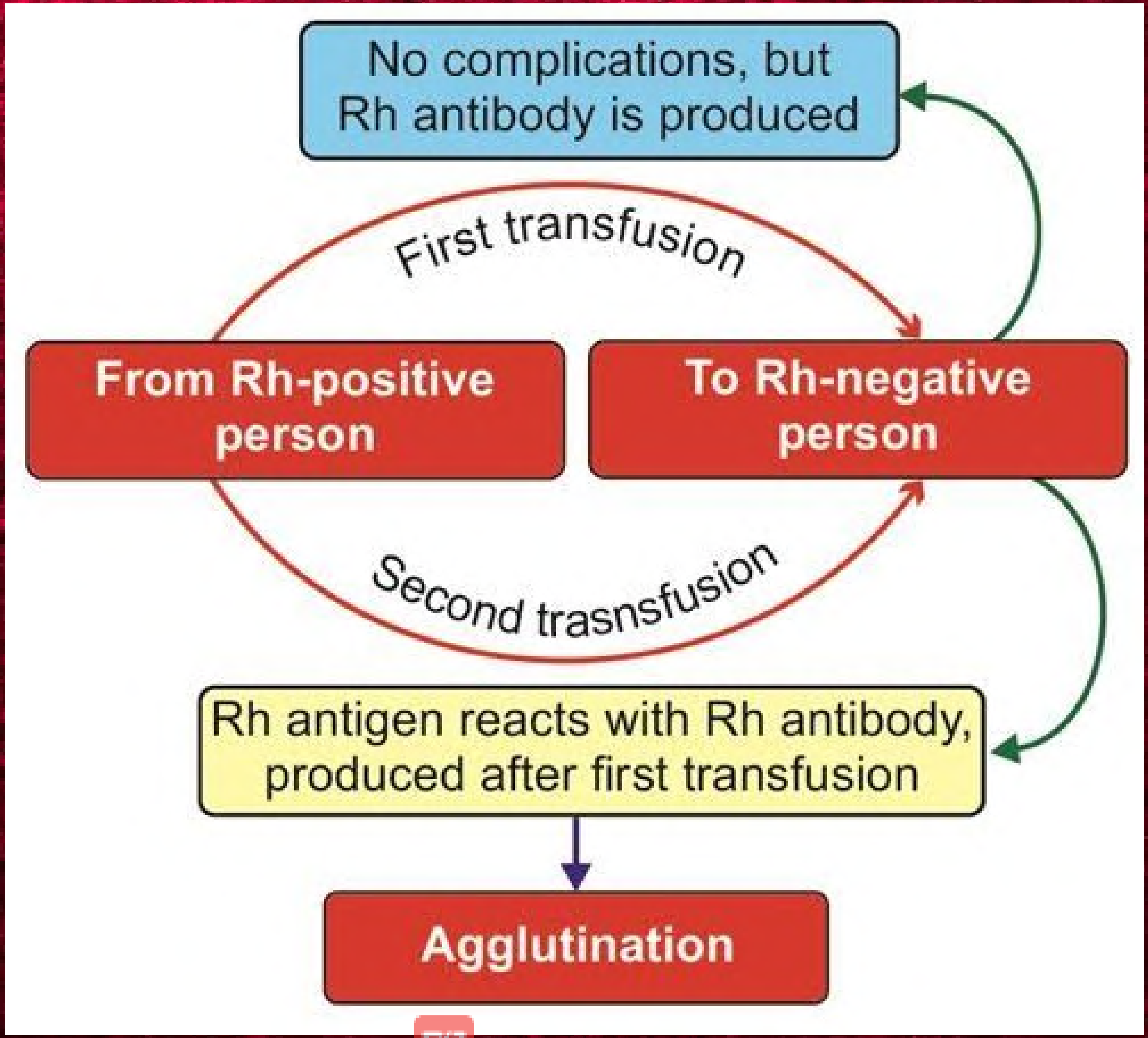
- Hemolytic disease is the disease in fetus and newborn, characterized by abnormal hemolysis of RBCs.
- It is due to **Rh incompatibility**, i.e. the difference between the Rh blood group of the mother and baby.
- Hemolytic disease leads to erythroblastosis fetalis.


Rh⁻
mother
Placenta
First
Rh⁺
fetus
Rh⁺
antigens
Anti-Rh
antibodies
Second
Rh⁺
fetus
(a) First
pregnancy
(b) Between
pregnancies
(c) Second
pregnancy

- Excessive haemolysis leads to severe complications ;

1. **Severe anemia**
2. **Hydrops fetalis**: edema , enlargement of liver & spleen, etc in fetus --> then death
3. **Kernicterus :** Brain damage caused by severe jaundice/ bilirubin encephalopathy

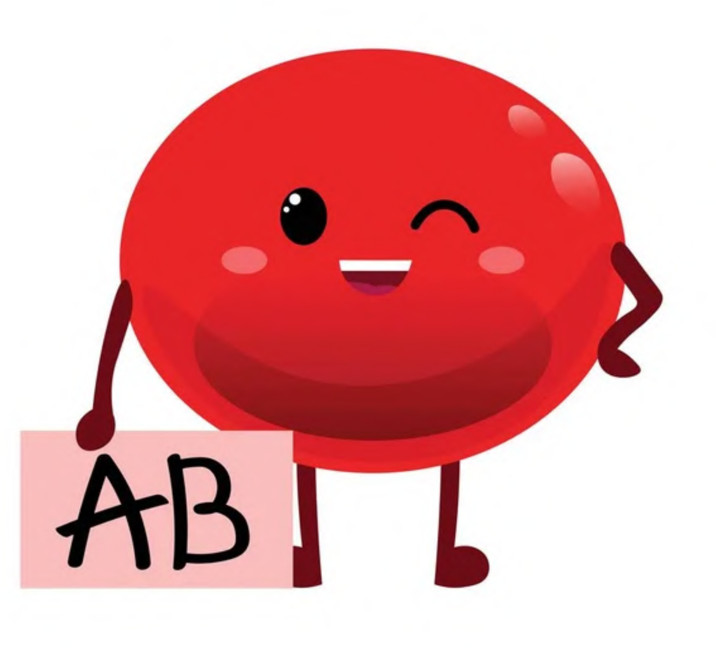
Rh Incompatibility



IMPORTANCE OF ABO GROUPS IN BLOOD TRANSFUSION

- During blood transfusion, only compatible blood must be used.
- The one who gives blood is called the '**donor**' and the one who receives the blood is called '**recipient**'.
- While transfusing the blood, **antigen of the donor** and the **antibody of the recipient** are considered.
- The antibody of the donor and antigen of the recipient are ignored mostly.

Universal Donors



Universal Recipient

Recipient	Blood donor			
	O	A	B	AB
O	✓	x	x	x
A	✓	✓	x	x
B	✓	x	✓	x
AB	✓	✓	✓	✓

BLOOD TRANSFUSION REACTION

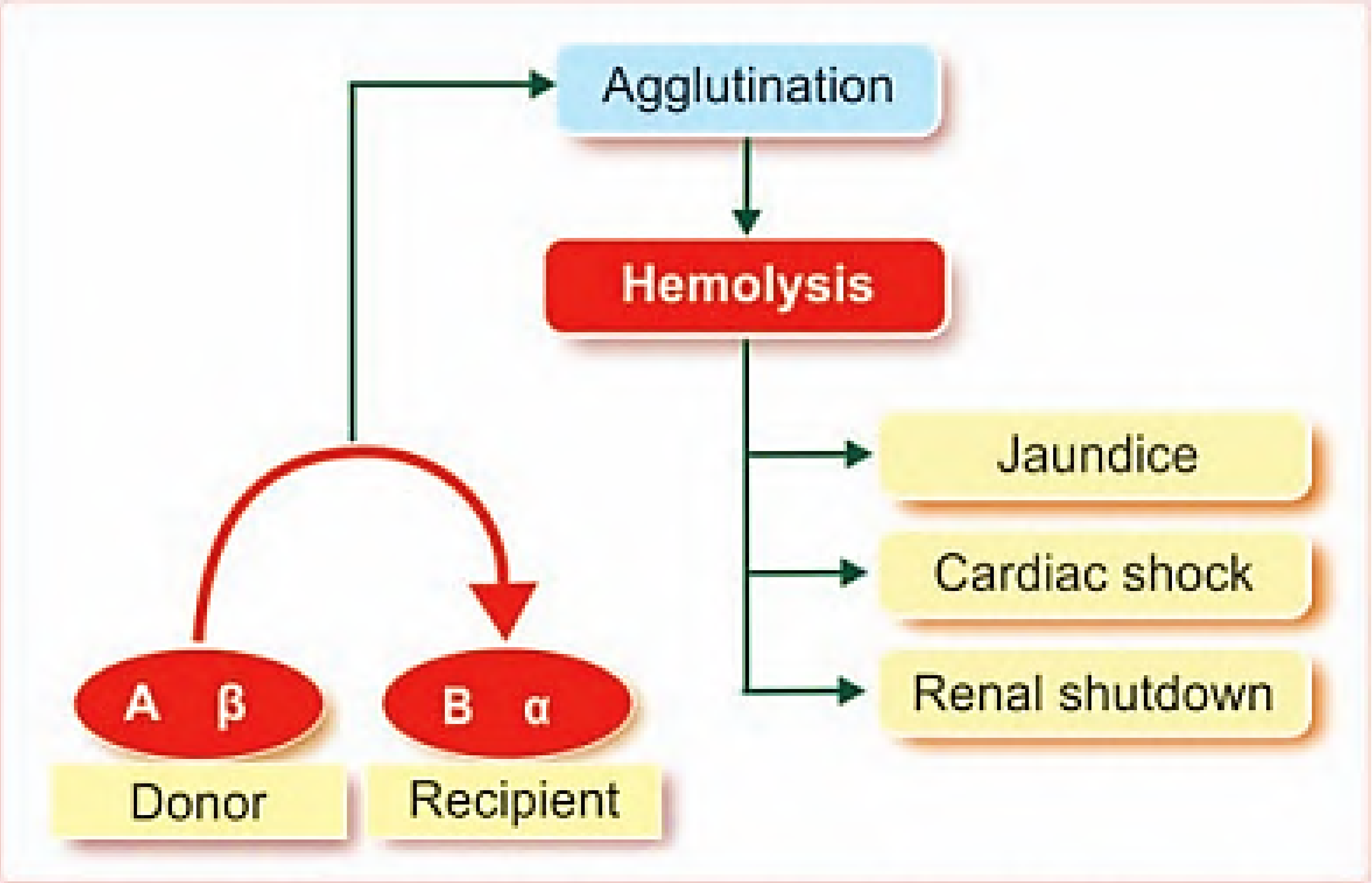
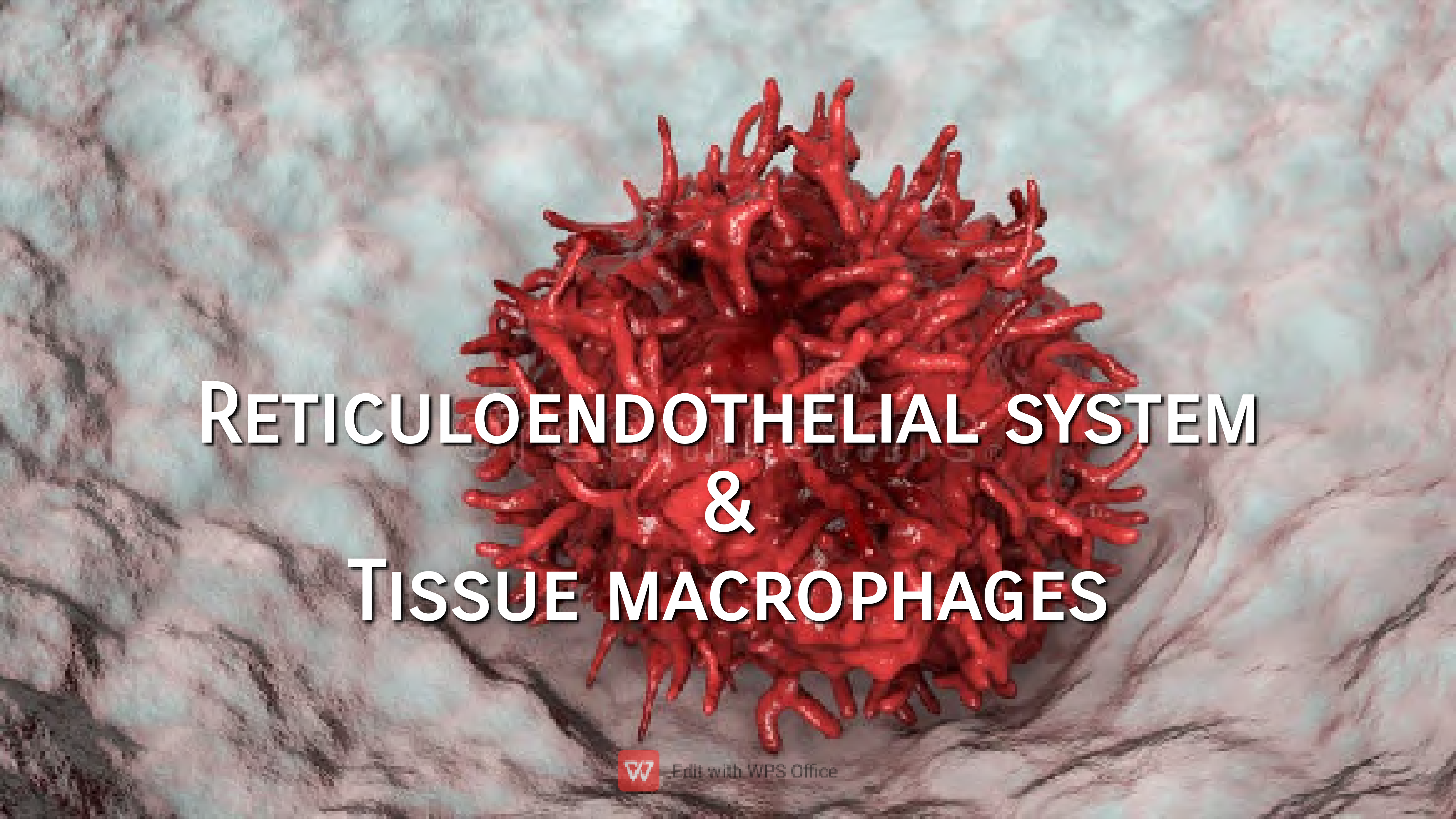


FIGURE 21.2: Complications of mismatched blood transfusion

Reticuloendothelial system & Tissue macrophages



- Reticuloendothelial system or tissue macrophage system is the **system of primitive phagocytic cells**, which play an important role in defense mechanism of the body.
- The reticuloendothelial cells are found in the following structures:
  1. Endothelial lining of vascular and lymph channels.
  2. Connective tissue and some organs like spleen, liver, lungs, lymph nodes, bone marrow, etc.

- Reticular cells in these tissues form the tissue macrophage system.
- **Macrophage is a large phagocytic cell, derived from monocyte**

CLASSIFICATION OF RETICULOENDOTHELIAL CELLS

- Two types:

1. **Fixed reticuloendothelial cells / tissue macrophages.**
2. **Wandering reticuloendothelial cells**

Fixed reticuloendothelial cells:

- Also called **fxed histiocytes** bcz, these cells are usually located in the tissues.

- Tissue macrophages are present in the following areas:

1. Connective tissues
2. Liver (**Kupffer cells**), spleen, bone marrow, adrenal glands ,etc.
3. **Meningocytes** in meninges.
4. **Tissue macrophages** present in lungs
5. Subcutaneous tissues.

Wandering reticuloendothelial cells and tissue macrophages:

- Also called free histiocytes.
- Two types of wandering reticuloendothelial cells:

1. **Free Histiocytes of Blood**
   1. Neutrophils
   2. Monocytes
2. **Free Histiocytes of Solid Tissue**: During emergency, the fixed histiocytes from connective tissue and other organs become wandering cells and enter the circulation

Functions of reticuloendothelial system

1. **Phagocytosis: Bacterial, dead cells, foreign particles (direct).**

- Phagocytosis is the processes of engulfing and ingestion of bacteria or other foreign bodies. It is part of the natural or innate immune process.
- Macrophages are a powerful phagocytic cells:
  a. Macrophages also means "**Big eater**", they are capable of phagocytosis.
  b. They are modified monocyte in tissues.
  c. Ingest up to 100 bacteria.
  d. Ingest larger particles such as old RBC.
  e. Get rid of waste products.

2. Secretion of Bactericidal Agents:

- Tissue macrophages secrete many bactericidal agents which kill the bacteria.
- The important bactericidal agents of macrophages are the **oxidants**. An oxidant is a substance that oxidizes another substance.
- Oxidants secreted by macrophages are:

i. **Superoxide (O_2^-)**

ii. **Hydrogen peroxide (H_2O_2)**

iii. **Hydroxyl ions (OH^-).** These oxidants are the most potent bactericidal agents.

So, even the bacteria which cannot be digested by lysosomal enzymes are degraded by these oxidants.

3. Secretion of Interleukins:

- Tissue macrophages secrete the following interleukins, which help in immunity:

i. **Interleukin-1 (IL-1):** Accelerates the maturation and proliferation of specific **B lymphocytes and T lymphocytes.**

ii. **Interleukin-6 (IL-6):** Causes the **growth of B lymphocytes and production of antibodies.**

iii. **Interleukin-12 (IL-12):** Influences the **T helper cells.**

4. Secretion of Tumor Necrosis Factors:

i. **TNF-α**: Causes necrosis of tumor and activates the immune responses in the body

ii. **TNF-β**: Stimulates immune system and vascular response, in addition to causing necrosis of tumor

5. Secretion of Transforming Growth Factor:

- Helps in **preventing rejection of transplanted tissues** or organs by immunosuppression.

6. Secretion of Colony-stimulation Factor:

- Colony-stimulation factor (CSF) secreted by macrophages is **M-CSF**. It **accelerates the growth of granulocytes, monocytes and macrophages.**

7. Secretion of Platelet-derived Growth Factor:

- which **accelerates repair of damaged blood vessel and wound healing.**

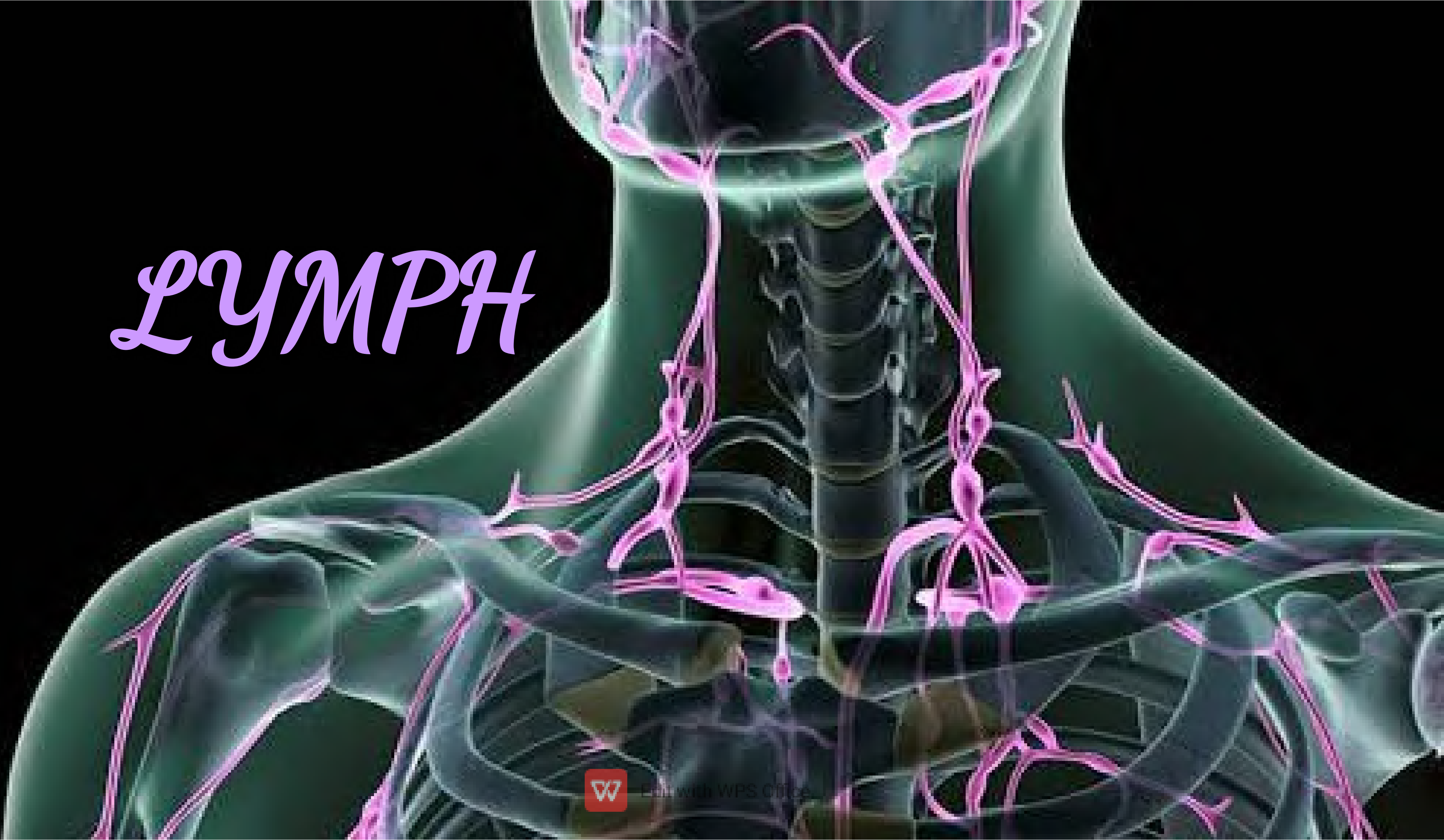
8. Removal of Carbon Particles and Silicon by ingestion

9. Destruction of Senile RBC:

10. Destruction of Hemoglobin:

- Hemoglobin released from broken senile RBCs is degraded by the reticuloendothelial cells.

LYMPH



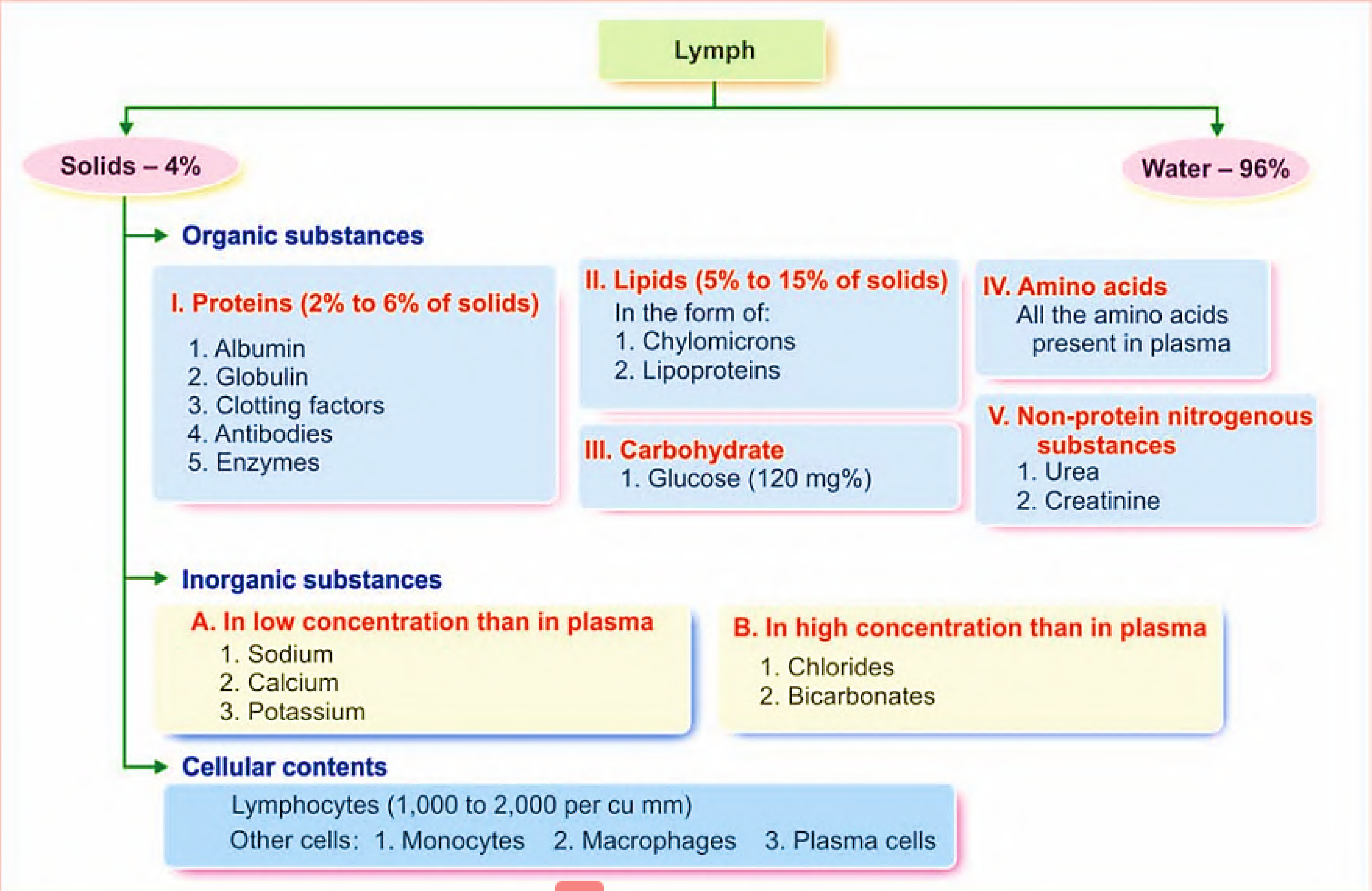


FIGURE 26.3: Composition of lymph

- Meaning:-After blood travels through capillary beds and is moved to the venous system, some of its fluid is left behind in the tissues it called lymph.

- Lymph is a **clear, colorless liquid** with a composition similar to blood plasma.

- It is nothing but the **clear, watery blood plasma leaked out through the capillary walls to flow around the cells.**

- It contains oxygen, proteins, glucose and white blood cells

- All the cells bathed in this fluid receive their nutrients and oxygen from it.

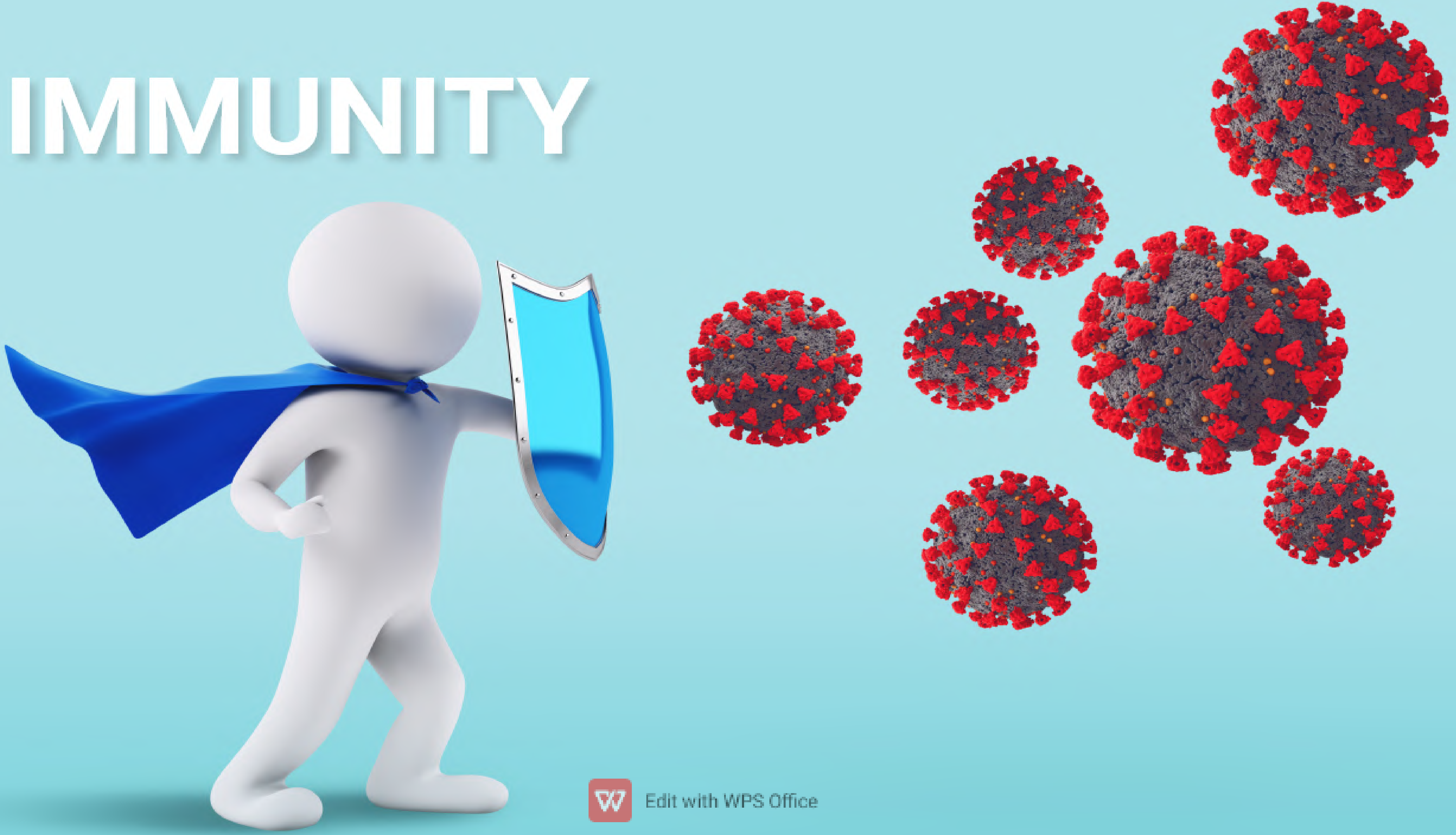
- Tubular vessels transport lymph back to the blood, ultimately replacing the volume lost during the formation of the interstitial fluid.

- Lymph movement occurs despite low pressure due to peristalsis, valves, and compression during contraction of adjacent skeletal muscle and arterial pulsation

Functions of lymph

1. Important function of lymph is to **return the proteins from tissue spaces into blood.**
2. **Redistribution of fluid** in the body.
3. Bacteria, toxins and other foreign bodies are removed from tissues via lymph.
4. Maintenance of **structural and functional integrity of tissue**. Obstruction to lymph flow affects various tissues, particularly myocardium, nephrons and hepatic cells.
5. Important route for **intestinal fat absorption**. This is why lymph appears milky after a fatty meal.
6. Role in immunity by transport of lymphocytes

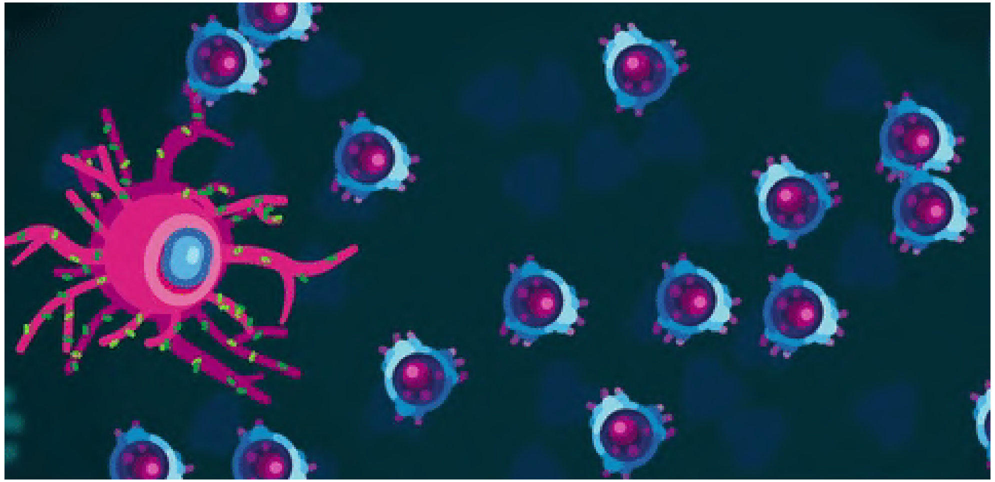
IMMUNITY

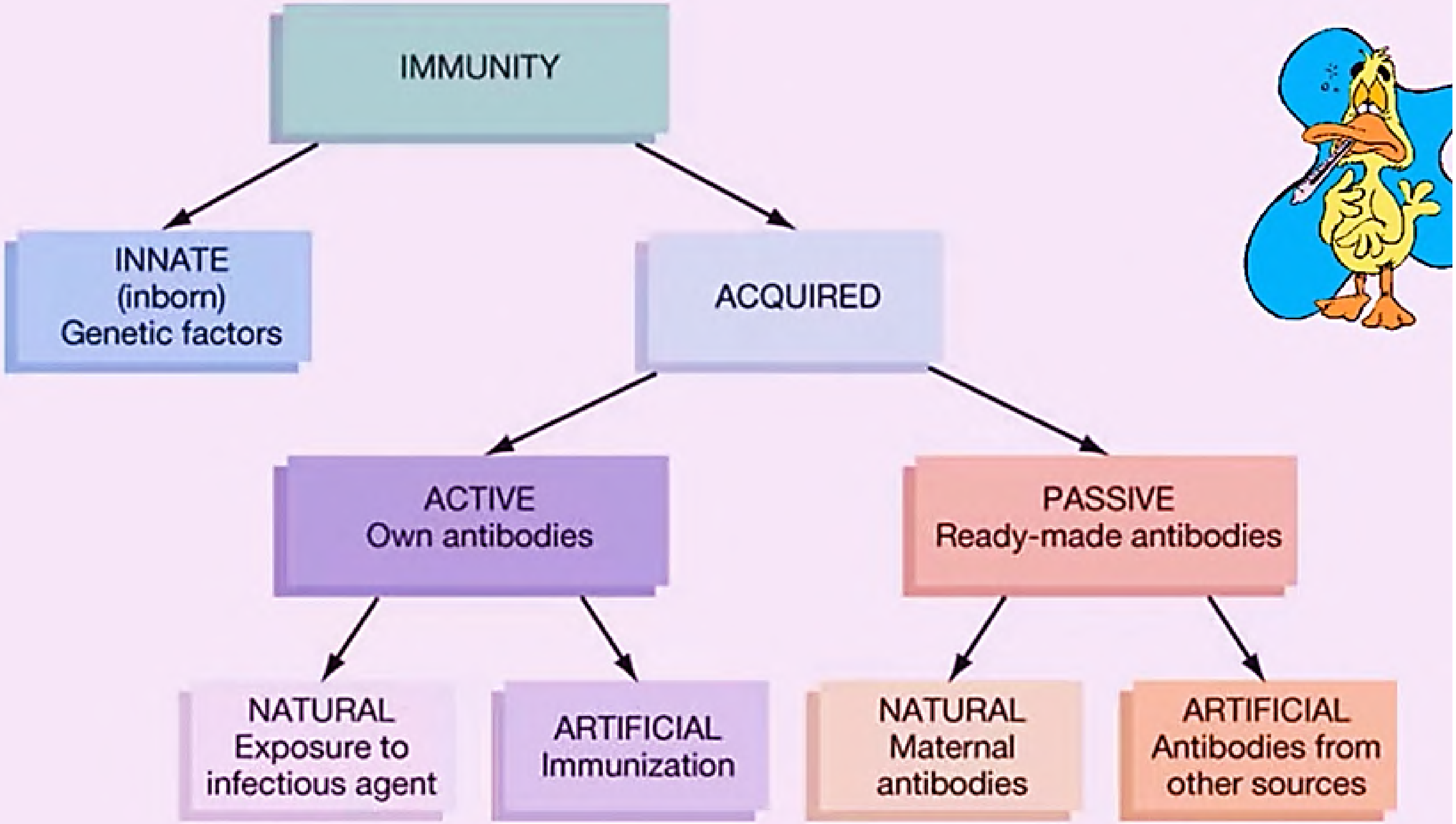


- Immunity is defined as the capacity of the body to resist pathogenic agents.

- It is the ability of body to resist the entry of different types of foreign bodies like bacteria, virus, toxic substances, etc.

- Immunity is of two types:
  I. Innate immunity.
  II. Acquired immunity




IMMUNITY
INNATE
(inborn)
Genetic factors
ACQUIRED
ACTIVE
Own antibodies
PASSIVE
Ready-made antibodies
NATURAL
Exposure to
infectious agent
ARTIFICIAL
Immunization
NATURAL
Maternal
antibodies
ARTIFICIAL
Antibodies from
other sources

Innate immunity / non-specific immunity:

- Innate immunity is the **inborn** capacity of the body to resist pathogens.
- By chance, if the organisms enter the body, innate immunity eliminates them before the development of any disease.
- It is otherwise called the natural or non-specific immunity.
- This type of immunity represents the **first line of defense** against any type of pathogens. Therefore, it is also called non-specific immunity.

Acquired immunity / specific immunity:

- Resistance developed in the body **against any specific foreign body** like bacteria, viruses, toxins, vaccines or transplanted tissues.
- So, this type of immunity is also known as specific immunity. It is the **most powerful immune mechanism** that protects the body from the invading organisms or toxic substances.
- Lymphocytes are responsible for acquired immunity.
- Types of Acquired Immunity :-

1. Cellular immunity
2. Humoral immunity.

- There are two types of lymphocytes: T-lymphocytes & B -lymphocytes.
- T–lymphocytes are involved in cellular immunity and B-lymphocytes are for humoral immunity.
- **B–lymphocytes** are of two types: **plasma cells & memory cells**
- **T-lymphocytes** are of two types: **Killer T cells or cytotoxic T cells, Memory T cells and Helper T cells**

CELL MEDIATED IMMUNITY:

- Type of immunity in which the **cells directly act on the antigens** to destroy them, **without the involvement of antibodies.**

- It is responsible for **delayed allergic reactions** and **rejection of transplants of foreign tissues.**

- **T – lymphocytes, macrophages and natural killer cells** are involved in cell – mediated immunity.

T – lymphocytes are of 4 types:

- **T – helper cells**
- **T – cytotoxic cells**
- **T – suppressor cells**
- **T – plasma cells**

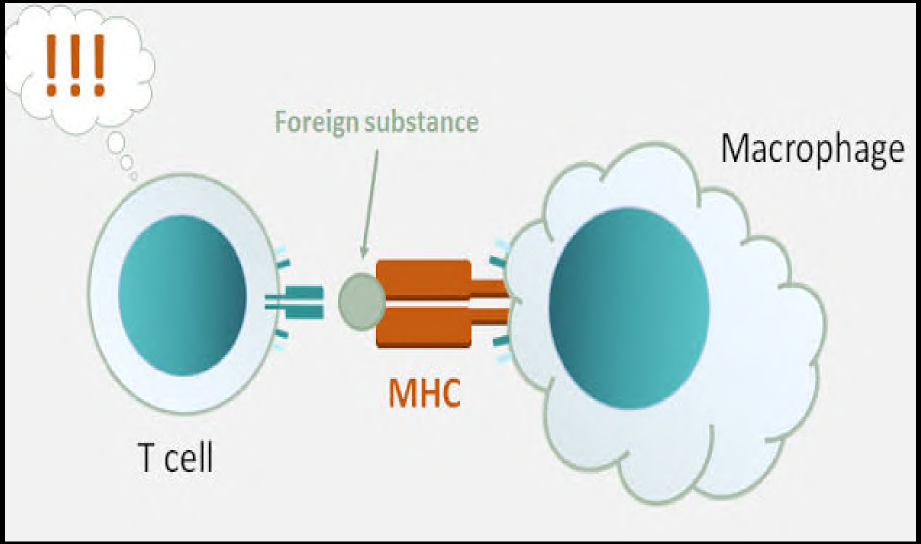
Antigen-presenting cells:

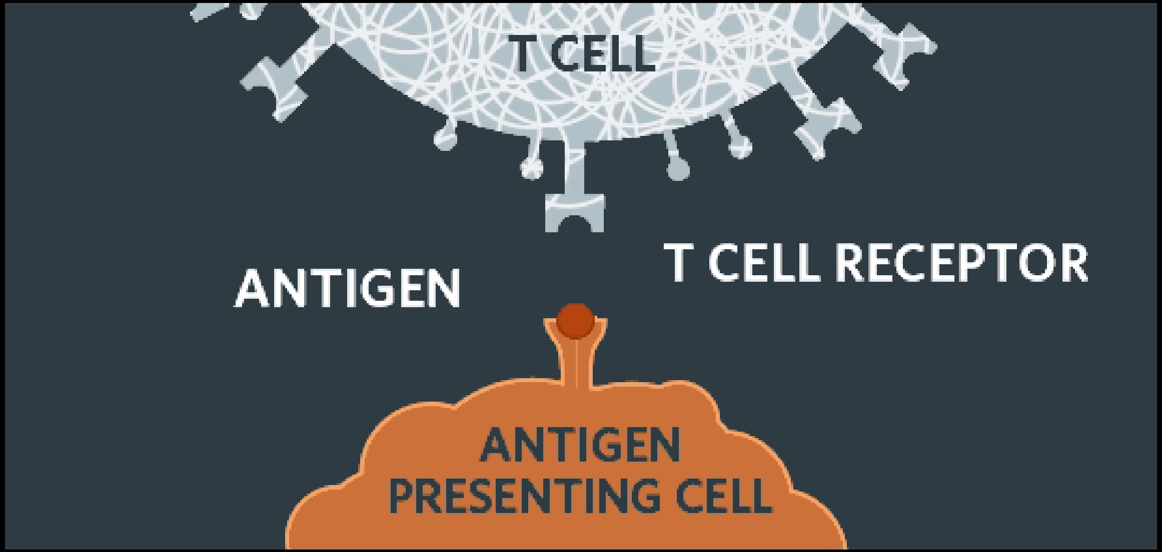
- Special type of cells in the body, which induce the release of antigenic materials from invading organisms and later present these materials to the helper T cells.
- Types of Antigen-Presenting Cells :-

1. **Macrophages**
2. **Dendritic cells**
3. **B lymphocytes.**

Role of Antigen-presenting Cells:

- **Invading foreign organisms** are either engulfed by **macrophages** through **phagocytosis or trapped by dendritic cells.**
- Later, the antigen from these organisms is **digested into small peptide products.**
- These antigenic peptide products **move towards the surface of the antigen-presenting cells** and **bind with human leukocyte antigen (HLA).**
- HLA is a genetic matter present in the molecule of **class II major histocompatibility complex (MHC)**, which is situated on the surface of the antigen presenting cells.
- B-cells ingest the foreign bodies **by means of pinocytosis.**

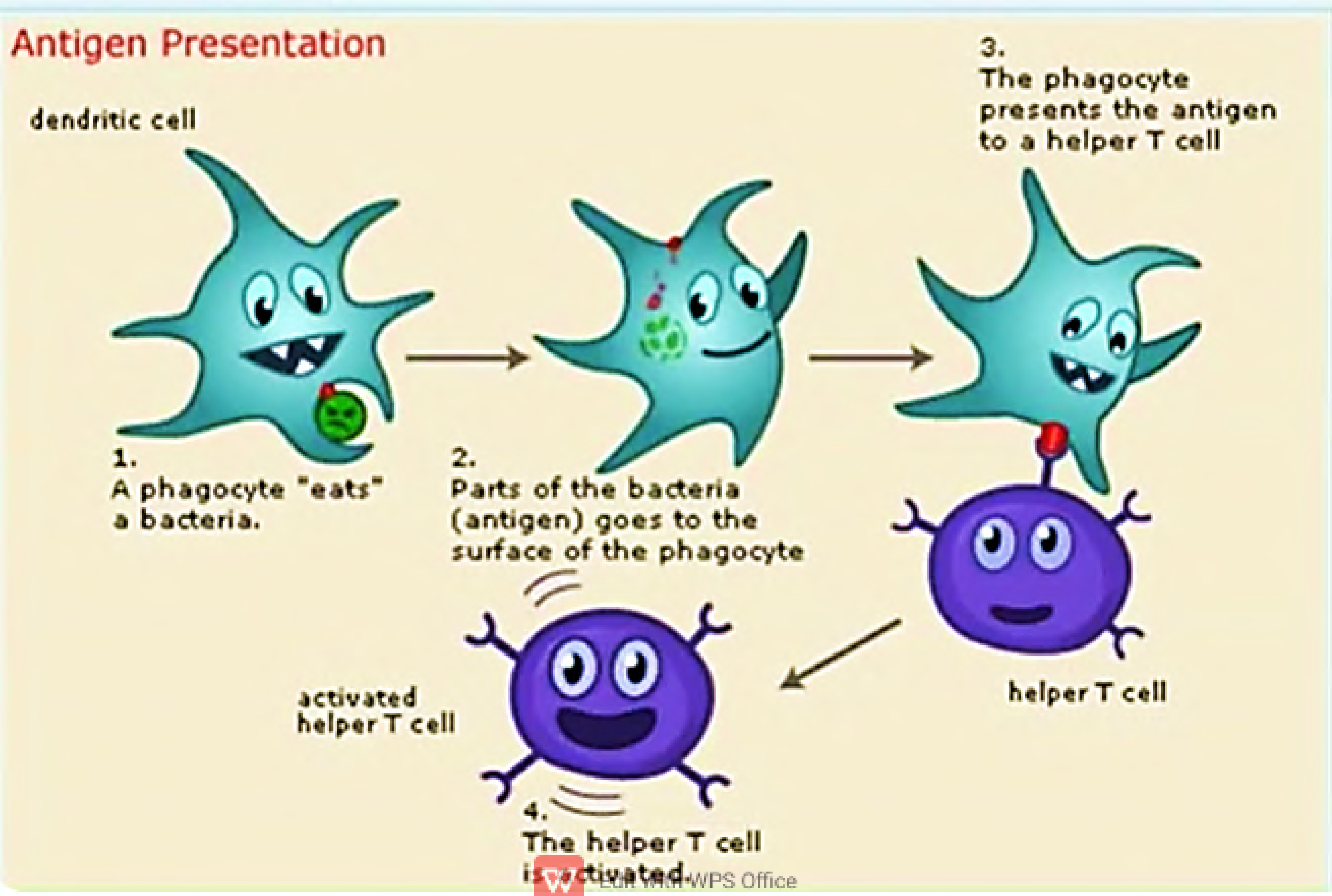

!!!
Foreign substance
Macrophage
MHC
T cell


T CELL
ANTIGEN
T CELL RECEPTOR
ANTIGEN
PRESENTING CELL

Presentation of Antigen:

- Antigen-presenting cells present their class II MHC molecules together with antigen-bound HLA to the **helper T cells.**

- This activates the helper T cells through series of events


Antigen Presentation
dendritic cell
1.
A phagocyte "eats"
a bacteria.
2.
Parts of the bacteria
(antigen) goes to the
surface of the phagocyte
3.
The phagocyte
presents the antigen
to a helper T cell
helper T cell
activated
helper T cell
4.
The helper T cell
is activated.

Role of helper T cells:

- Helper T cells (CD4 cells) which **enter the circulation activate all the other T cells and B cells.**
- Normal, CD4 count in healthy adults varies between 500 and 1500 per cubic millimeter of blood.
- Helper T cells are of two types:
  1. Helper-1 (TH1) cells
  2. Helper-2 (TH2) cells

Role of cytotoxic T cells:

- Cytotoxic T cells that are activated by helper T cells, circulate through blood, lymph and lymphatic tissues and destroy the invading organisms by attacking them directly.

- Mechanism of Action of Cytotoxic T Cells:

1. Receptors situated on the outer membrane of cytotoxic T cells bind the antigens or organisms tightly with cytotoxic T cells.
2. Then, the cytotoxic T cells enlarge and release cytotoxic substances like the lysosomal enzymes.
3. These substances destroy the invading organisms.
4. Like this, each cytotoxic T cell can destroy a large number of microorganisms one after another

Role of suppressor T cells:

- Suppressor T cells are also called **regulatory T cells.**
- These T cells suppress the activities of the killer T cells.
- Thus, the suppressor T cells play an important role in **preventing the killer T cells from destroying the body's own tissues along with invaded organisms.**
- Suppressor cells suppress the activities of helper T cells also.

Role of memory T cells:

- **Some of the T cells activated by an antigen do not enter the circulation but remain in lymphoid tissue.**
- These T cells are called **memory T cells.**
- In later periods, the memory cells migrate to various lymphoid tissues throughout the body.
- When the body is exposed to the same organism for the second time, the memory cells identify the organism and immediately activate the other T cells.
- So, the invading organism is destroyed very quickly. The response of the T cells is also more powerful this time.

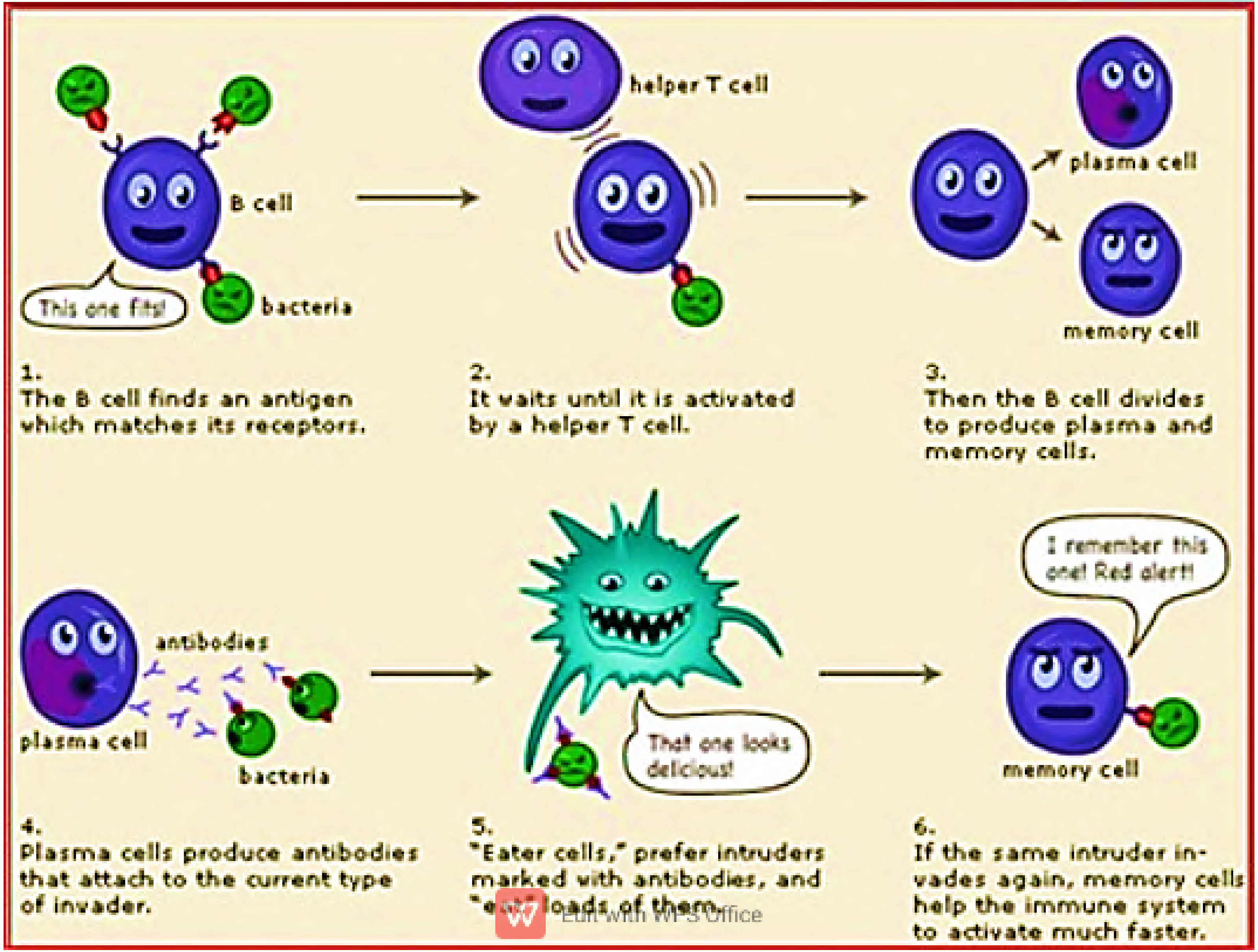
Specificity of T cells:

- Each T cell is designed to be activated only by one type of antigen.
- It is capable of developing immunity against that antigen only.
- This property is called the specificity of T cells.

HUMORAL IMMUNITY

- Type of immunity in which **antibodies** are produced against the antigens attacking the body tissue.
- **B – lymphocytes** are responsible for this type of immunity.
- 2 types:

1. **Plasma cells**
2. **memory cells**


helper T cell
B cell
This one fits!
bacteria
plasma cell
memory cell
1.
The B cell finds an antigen which matches its receptors.
2.
It waits until it is activated by a helper T cell.
3.
Then the B cell divides to produce plasma and memory cells.
antibodies
plasma cell
bacteria
That one looks delicious!
I remember this one! Red alert!
memory cell
4.
Plasma cells produce antibodies that attach to the current type of invader.
5.
"Eater cells," prefer intruders marked with antibodies, and "eat" loads of them.
6.
If the same intruder invades again, memory cells help the immune system to activate much faster.

Role of antigen-presenting cells:

- The ingestion of foreign organisms and digestion of their antigen by the antigen-presenting cells occurs.

- Antigen-presenting cells present the antigenic products bound with HLA (which is present in class II MHC molecule) to B cells.
- This activates the B cells through series of events.

Role of plasma cells:

- Plasma cells destroy the foreign organisms by **producing the antibodies.**
- Antibodies are globulin in nature.
- The rate of the antibody production is very high, i.e. each plasma cell produces about 2000 molecules of antibodies per second.
- The antibodies are also called immunoglobulins.
- Antibodies are released into lymph and then transported into the circulation.
- The antibodies are produced until the end of lifespan of each plasma cell, which may be from several days to several weeks

B-cell

Role of memory B cells:

- Memory B cells occupy the lymphoid tissues throughout the body.
- The memory cells are in inactive condition until the body is exposed to the same organism for the second time.
- During the second exposure, the memory cells are stimulated by the antigen and produce more quantity of antibodies at a faster rate, than in the first exposure.
- The antibodies produced during the second exposure to the foreign antigen are also more potent than those produced during first exposure.
- This phenomenon forms the basic principle of vaccination against the infections.

Role of helper t cells:

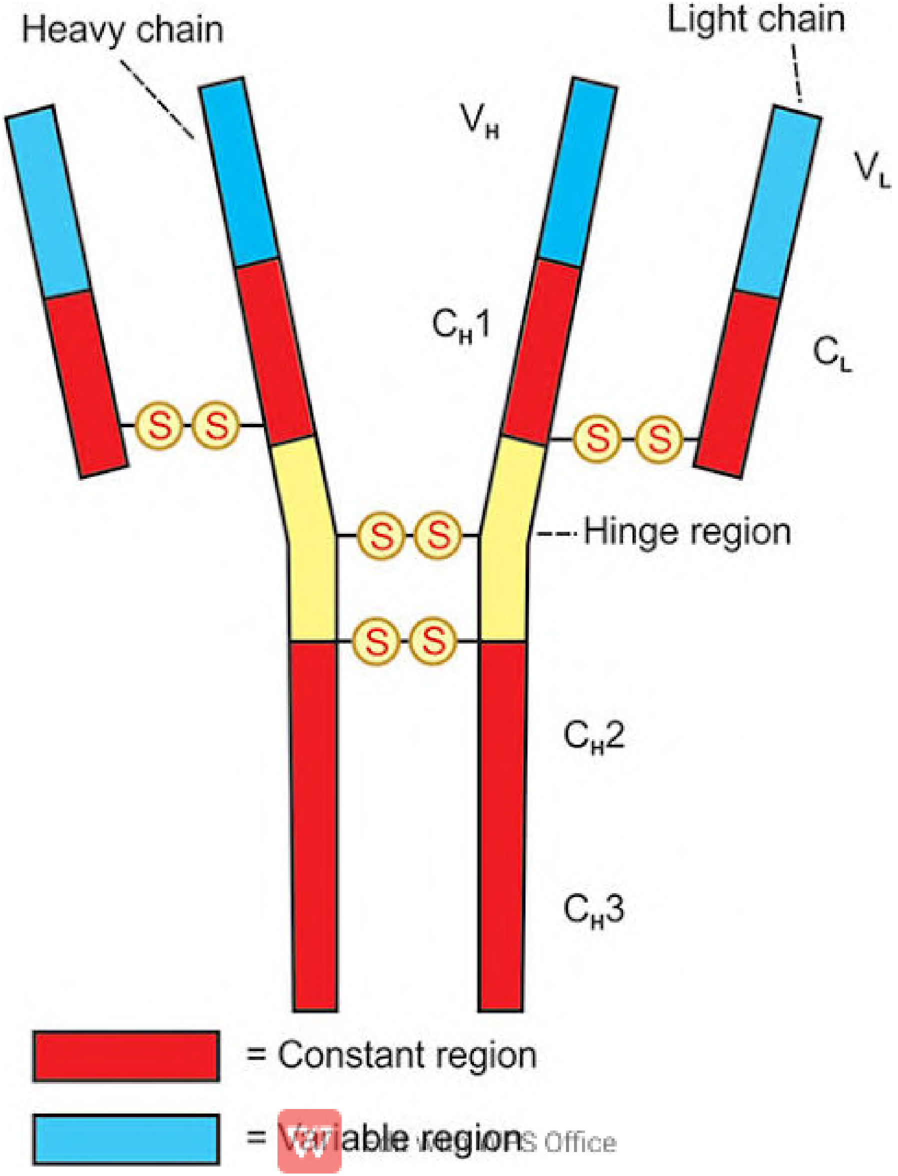
- Helper T cells are simultaneously activated by antigen.

- Activated helper T cells secrete two substances called **interleukin-2 and B cell growth factor,** which promote:

1. **Activation of more number of B lymphocytes.**
2. **Proliferation of plasma cells.**
3. **Production of antibodies**

Antibodies or immunoglobulins:

- An antibody is defined as a protein that is produced by B lymphocytes in response to the presence of an antigen.
- Antibody is gamma globulin in nature and it is also called immunoglobulin (Ig).
- Immunoglobulins form 20% of the total plasma proteins.
- Antibodies enter almost all the tissues of the body.


Heavy chain
Light chain
V_H
V_L
C_H1
C_L
S S
S S
S S
S S
Hinge region
C_H2
C_H3
= Constant region
= Variable region

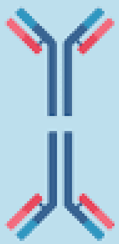
- Types of Antibodies Five types of antibodies are identified:

1. IgG (Ig gamma) -Most abundant
2. IgA (Ig alpha) - About 10 - 15%
3. IgM (Ig mu) - 5 - 10%
4. IgE (Ig epsilon) - Less than 0.1%
5. IgD (Ig delta) - 0.2%

- Among these antibodies, IgG forms 75% of the antibodies in the body.
- Only IgG can cross placenta

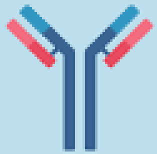
5 Types of Antibodies

Antibodies or immunoglobulins (Ig) are Y-shaped proteins that recognize unique markers (antigens) on pathogens.



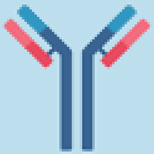
IgA

Secreted into mucous, saliva, tears, colostrum. Tags pathogens for destruction.



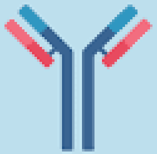
IgD

B-cell receptor. Stimulates release of IgM.



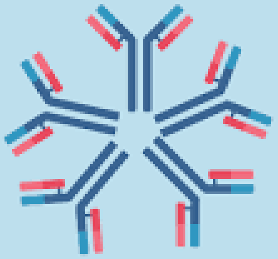
IgE

Binds to mast cells and basophils. Allergy and antiparasitic activity.



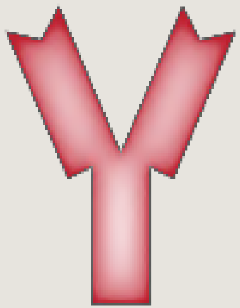
IgG

Binds to phagocytes. Main blood antibody for secondary responses. Crosses placenta.

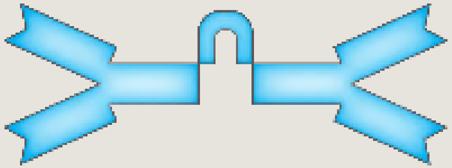


IgM

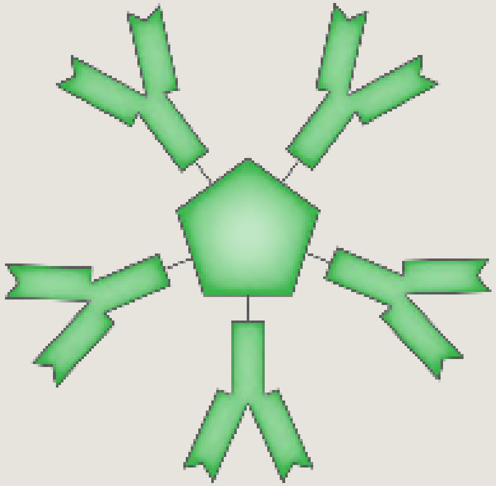
Fixes complement. Main antibody of primary responses. B-cell receptor. Immune system memory.



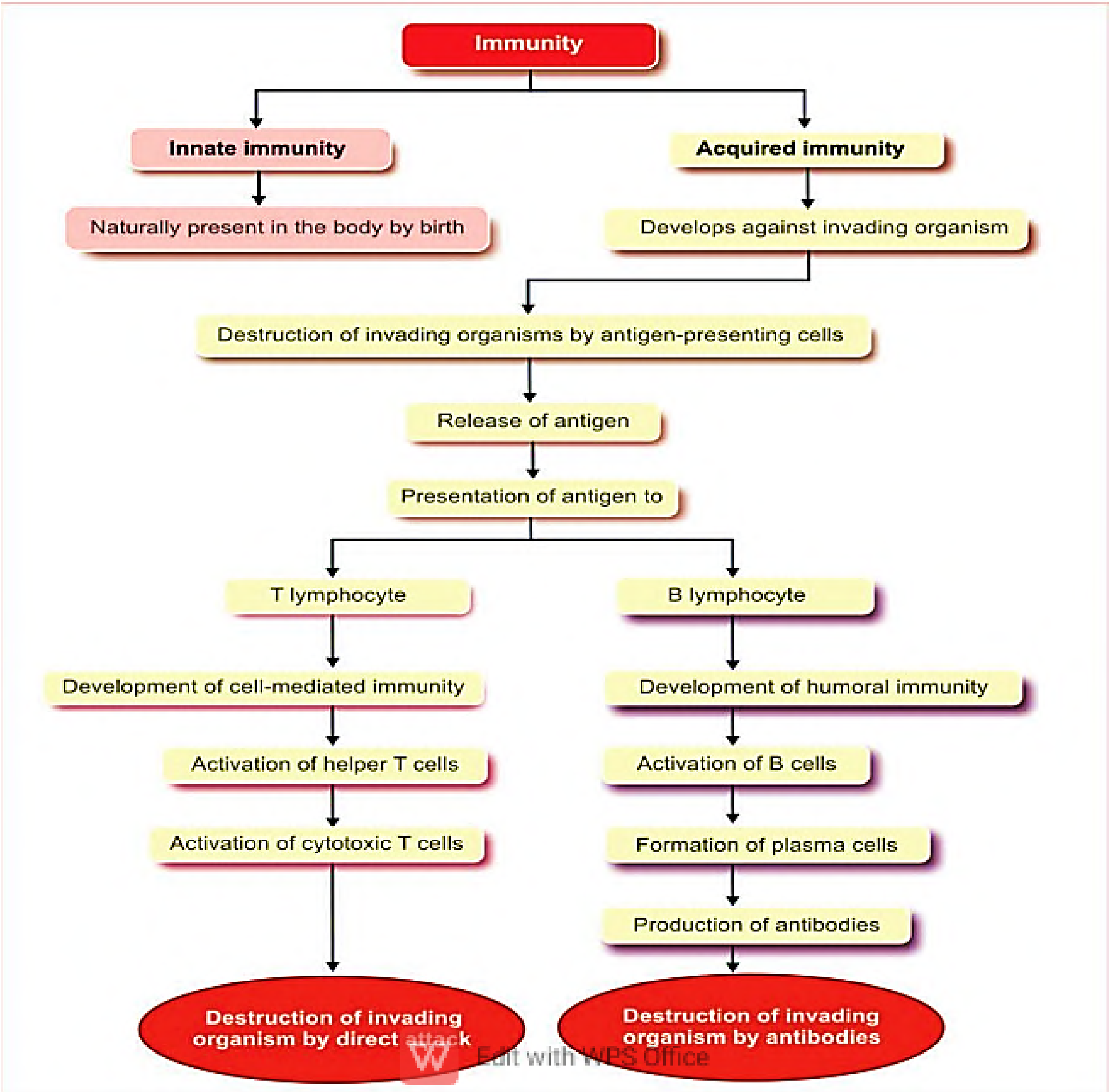
Monomer
IgD, IgE, IgG



Dimer
IgA



Pentamer
IgM


Immunity
Innate immunity
Acquired immunity
Naturally present in the body by birth
Develops against invading organism
Destruction of invading organisms by antigen-presenting cells
Release of antigen
Presentation of antigen to
T lymphocyte
B lymphocyte
Development of cell-mediated immunity
Development of humoral immunity
Activation of helper T cells
Activation of B cells
Activation of cytotoxic T cells
Formation of plasma cells
Production of antibodies
Destruction of invading organism by direct attack
Destruction of invading organism by antibodies

NATURAL KILLER CELL

- It is a large granular cell that plays an important role in defense mechanism of the body.
- It has an indented nucleus.
- Considered as the third type of lymphocyte.
- It is derived from bone marrow.
- NK cell is said to be the **first line of defense** in specific immunity, particularly against viruses.

- NK cell kills the invading organisms or the cells of the body without prior sensitization.

- It is not a phagocytic cell but its granules contain hydrolytic enzymes such as **perforins and granzymes.**

- These hydrolytic enzymes play an important role in the lysis of cells of invading organisms.

- Functions of Natural Killer (NK) Cell Natural killer cell:

1. **Destroys the viruses**
2. **Destroys the viral infected or damaged cells, which might form tumors**
3. **Destroys the malignant cells and prevents development of cancerous tumors**
4. **Secretes cytokines such as interleukin-2, interferons, colony stimulating factor (GM-CSF) and tumor necrosis factor-α.**

CYTOKINES

- Cytokines are the hormone-like small proteins acting as intercellular messengers (**cell signaling molecules**) by binding to specific receptors of target cells.
- These **non-antibody proteins are secreted by WBCs and some other types of cells.**
- Their major function is the activation and regulation of general immune system of the body.
- Cytokines are distinct from the other cell-signaling molecules such as growth factors and hormones.

IMMUNIZATION

- Immunization is defined as the procedure by which the body is prepared to fight against a specific disease.
- It is used to induce the immune resistance of the body to a specific disease.
- Immunization is of two types:

1. **Passive immunization**
2. **Active immunization**

Passive immunization

- Immunity is produced without challenging the immune system of the body.

- It is done by **administration of serum or gamma globulins from a person who is already immunized (affected by the disease) to a non-immune person.**

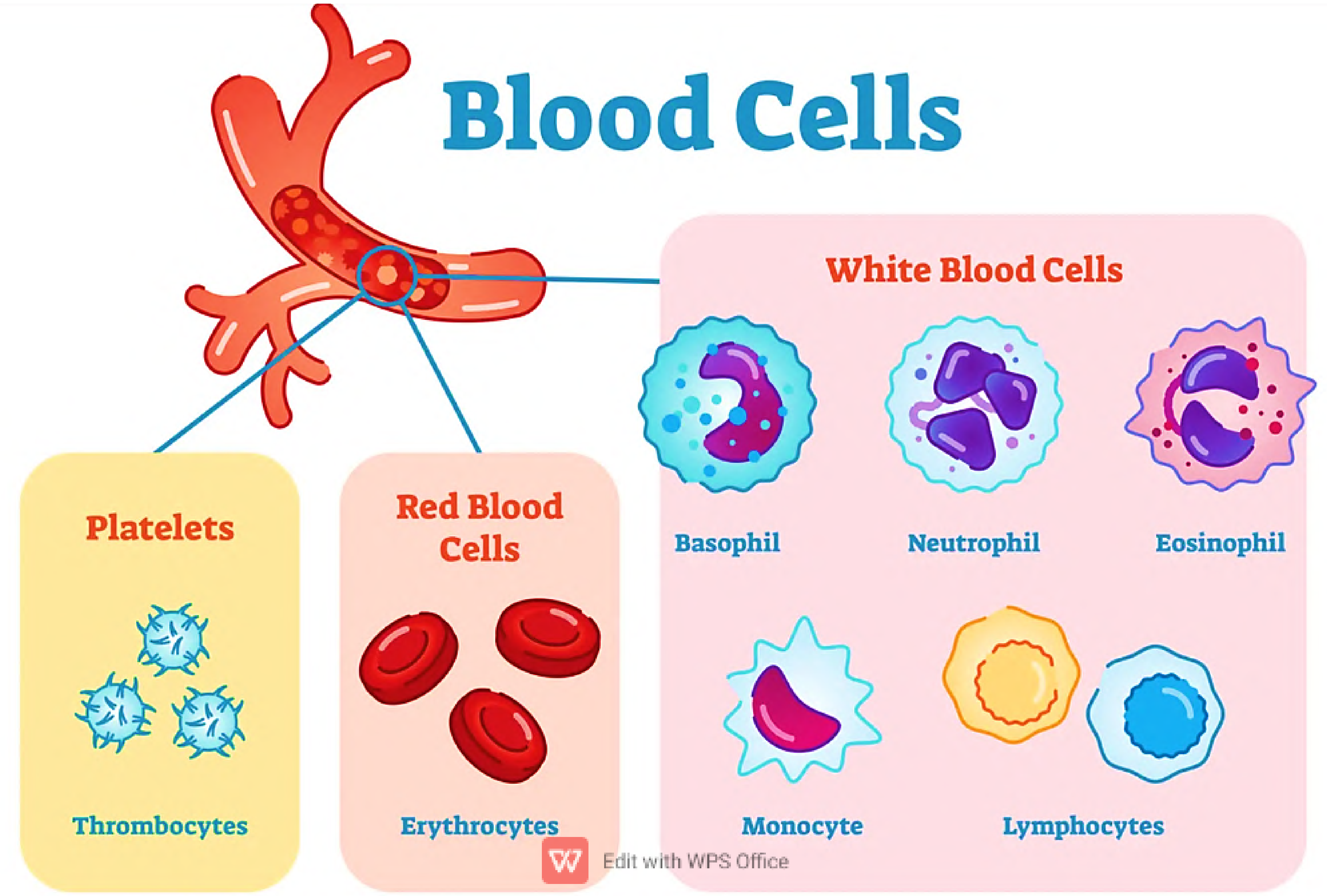
- Passive immunization is acquired either naturally or artificially

Active immunization

- immunity is acquired by activating immune system of the body.

- Body develops resistance against disease by producing antibodies **following the exposure to antigens.**

- Active immunity is acquired either naturally or artificially.


Blood Cells
White Blood Cells
Basophil
Neutrophil
Eosinophil
Monocyte
Lymphocytes
Platelets
Thrombocytes
Red Blood Cells
Erythrocytes


THANK
YOU!